Nonparametric Eigenvalue-Regularized Precision or Covariance Matrix Estimator

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Abstract

We introduce nonparametric regularization of the eigenvalues of a sample covariance matrix through splitting of the data (NERCOME), and prove that NERCOME enjoys asymptotic optimal nonlinear shrinkage of eigenvalues with respect to the Frobenius norm. One advantage of NERCOME is its computational speed when the dimension is not too large. We prove that NERCOME is positive definite almost surely, as long as the true covariance matrix is so, even when the dimension is larger than the sample size. With respect to the Stein's loss function, the inverse of our estimator is asymptotically the optimal precision matrix estimator, with optimality proved through what Ledoit and Wolf (2013a) has found. Asymptotic efficiency loss is defined through comparison with an ideal estimator, which assumed the knowledge of the true covariance matrix. We show that the asymptotic efficiency loss of NERCOME is almost surely 0 with a suitable split location of the data. We also show that all the aforementioned optimality holds for data with a factor structure. Our method avoids the need to first estimate any unknowns from a factor model, and directly gives the covariance or precision matrix estimator. We compare the performance of our estimators with other methods through extensive simulations and a real data analysis.

Key words and phrases. High dimensional data analysis; covariance matrix; Stieltjes transform; data splitting; nonlinear shrinkage; factor model.

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1 Introduction

Thanks to the rapid development of computing power and storage in recent years, large data sets are becoming more readily available. Analysis of large data sets for knowledge discovery thus becomes more important in various fields. One basic and important input in data analysis is the covariance matrix or its inverse, called the precision matrix. For a large data set, the high dimensionality of the data adds much difficulty for estimating these matrices. One of the main difficulty is having an ill-conditioned sample covariance matrix when the dimension p is large relative to the sample size n (Ledoit and Wolf, 2004).

Realizing the ill-conditioned nature of the sample covariance matrix, much efforts are devoted to imposing special structures in estimating the covariance or the precision matrix. It is hoped that the true underlying matrices indeed conform to these structures, so that convergence results can be proved. Examples include a banded structure (Bickel and Levina, 2008b), sparseness of the covariance matrix (Bickel and Levina, 2008a, Cai and Zhou, 2012, Lam and Fan, 2009, Rothman et al., 2009), sparseness of the precision matrix related to a graphical model (Friedman et al., 2008, Meinshausen and Bühlmann, 2006), sparseness related to the modified Cholesky decomposition of a covariance matrix (Pourahmadi, 2007), a spiked covariance matrix from a factor model (Fan et al., 2008, 2011), or combinations of them (see Fan et al. (2013) for example).

Without a particular structure assumed, Stein (1975) and lecture 4 of Stein (1986) proposed the use of the class of rotation-equivariant estimators that retains the eigenvectors of the sample covariance matrix, but shrinks its eigenvalues. Hence, the smaller eigenvalues are made larger, while the larger eigenvalues are made smaller. This proposal indeed makes perfect sense, since Bai and Yin (1993) provided solid theoretical justification that, when the dimension p grows with the sample size n, the extreme eigenvalues of the sample covariance matrix are more extreme than the population counterparts. Shrinkage of the eigenvalues of a sample covariance matrix then becomes an important branch for covariance matrix estimation. Ledoit and Wolf (2004) proposed a well-conditioned covariance matrix estimator based on a weighted average of the identity and the sample covariance matrix. In effect, it shrinks the eigenvalues towards their grand mean. Won et al. (2013) proposed a condition number regularized estimator, which has the middle portion of the sample eigenvalues unchanged, and the more extreme eigenvalues are winsorized at certain constants.

Recently, using random matrix theory, Ledoit and Wolf (2012) proposed a class of rotation-equivariant covariance estimator with nonlinear shrinkage of the eigenvalues. They demonstrate great finite sample performance of their estimator in difficult settings. Ledoit and Wolf (2013b) extends their results to allow for $p > n$, and proposes the estimation of the spectrum of a covariance matrix. At the same time, an independent idea of regularization is proposed in Abadir et al. (2014) for a class of rotation-equivariant estimator, where the data is split into two, and the eigenvalues are regularized by utilizing the two independent sets of split data. However, their theoretical analysis is based on the assumption that the dimension of the covariance matrix is fixed.

In this paper, we investigate the theoretical properties of the regularized eigenvalues in Abadir et al. (2014). As a first contribution, we show that these eigenvalues are in fact asymptotically the same as those nonlinearly shrunk ones in Ledoit and Wolf (2012), when the observations $\mathbf{y}_i$ can be written as $\mathbf{y}_i = \mathbf{\Sigma}_p^{1/2}\mathbf{z}_i$, with $\mathbf{\Sigma}_p$ being the true covariance matrix, and $\mathbf{z}_i$ a vector of independent and identically distributed entries (see section 2 and assumption (A1) therein for more details). The method for estimating the nonlinearly shrunk eigenvalues in Ledoit and Wolf (2012) involves nonconvex optimizations, and is solved via a commercial package. On the other hand, the method in Abadir et al. (2014) involves only eigen-decompositions of $p \times p$ matrices, and can be much quicker to do when p is not too large, while commercial packages are not required. This can be particularly attractive for practitioners.

As a second contribution, we also show that, if the data is from a factor model, while the low dimensional factor can render the nonlinear shrinkage formula in Ledoit and Wolf (2012) incorrect, the regularized eigenvalues using the data splitting idea from Abadir et al. (2014) are still asymptotically optimal, when we consider minimizing the Frobenius loss for estimating the covariance matrix, or the inverse Stein's loss for estimating the precision matrix (see sections 2.3, 2.4 and 3 for more details). It means that, in estimating the covariance matrix for data from a factor model, we do not need to explicitly estimate the number of factors, the factor loading matrix and the unknown factor series, which can be difficult tasks themselves.

As a third contribution, we define a notion of efficiency for our estimator through comparing it with an "ideal" one, which is also rotation-equivariant itself and assumed the knowledge of the true covariance matrix. We prove that, almost surely, our estimator does not lose any efficiency compared to such an ideal estimator asymptotically. We also show that our estimator is almost surely positive definite. All these proofs are under the high dimensional setting $p/n \to c > 0$, where n is the sample size. This provides an important relaxation to the assumptions in Abadir et al. (2014), where p is either fixed or $p/n \to 0$. See section 4 for more details.

The rest of the paper is organized as follows. We first present the framework for the data together with the notations and the main assumptions to be used in section 2. We then present the generalization to the Marčenko-Pastur law proposed in Ledoit and Péché (2011) in section 2.1, followed by the sample splitting regularization idea in Abadir et al. (2014) in section 2.2. Together, these inspire us to propose our covariance matrix estimator in section 2.3, where we also present a theorem to show that our estimator is asymptotically optimal in a certain sense. We also show that our precision matrix estimator is asymptotically optimal with respect to the inverse Stein's loss function in section 2.4. Extension of our results to data from a factor model is introduced in section 3. Efficiency loss of an estimator is defined in section 4, with asymptotic efficiency of our estimator shown. Improvement to finite sample properties, and the choice of the split of data for sample splitting are discussed in section 4. Real data analysis, and simulation results comparing with the performance of other state-of-the-art methods, are given in section 5. Finally, the conclusion is given in section 6. All the proofs of our results are given in the Appendix.

2 The Framework and Overview of Relevant Results

Let $\mathbf{y}_i$, $i = 1, \ldots, n$ be an independent and identically distributed sample that we can observe, where each $\mathbf{y}_i$ is of length p. In this paper, we assume that $p = p_n$, such that $p/n \to c > 0$ as $n \to \infty$. The subscript n will be dropped if no ambiguity arises.

We let $\mathbf{Y} = (\mathbf{y}_1, \ldots, \mathbf{y}_n)$, and define the sample covariance matrix as

$$\mathbf{S}_n = \frac{1}{n}\mathbf{Y}\mathbf{Y}^{\mathrm{T}}. \tag{2.1}$$

We assume $\mathbf{y}_i$ has mean $\mathbf{0}$ (the vector of zeros with suitable size), and the covariance matrix is denoted as $\mathbf{\Sigma}_p = E(\mathbf{y}_i\mathbf{y}_i^{\mathrm{T}})$ for each i. We assume the following for Theorem 1 in section 2.3 to hold. Related to asymptotic efficiency of our estimator, Theorem 5 in section 4 has more restrictive moment assumptions. See section 4 for more details.

(A1) Each observation can be written as $\mathbf{y}_i = \mathbf{\Sigma}_p^{1/2}\mathbf{z}_i$ for $i = 1, \ldots, n$, where each $\mathbf{z}_i$ is a $p \times 1$ vector of independent and identically distributed random variables z_{ij}. Each z_{ij} has mean 0 and unit variance, and $E|z_{ij}|^{2k} = O(p^{k/2-1})$, $k = 2, 3, 4, 5, 6$.

(A2) The population covariance matrix $\mathbf{\Sigma}_p$ is non-random and of size $p \times p$. Furthermore, $\|\mathbf{\Sigma}_p\| = O(p^{1/2})$, where $\|\cdot\|$ is the L_2 norm of a matrix.

(A3) Let $\tau_{n,1} \geq \ldots \geq \tau_{n,p}$ be the p eigenvalues of $\mathbf{\Sigma}_p$, with corresponding eigenvectors $\mathbf{v}_{n,1}, \ldots, \mathbf{v}_{n,p}$. Define $H_n(\tau) = p^{-1}\sum_{i=1}^{p}\mathbf{1}_{\{\tau_{n,i}\leq\tau\}}$ the empirical distribution function (e.d.f.) of the population eigenvalues, where $\mathbf{1}_A$ is the indicator function of the set A. We assume $H_n(\tau)$ converges to some non-random limit H at every point of continuity of H.

(A4) The support of H defined above is the union of a finite number of compact intervals bounded away from zero and infinity. Also, there exists a compact interval in $(0, +\infty)$ that contains the support of

H_n for each n.

These four assumptions are very similar to assumptions A1 to A4 in Ledoit and Wolf (2012) and (H_1) to (H_4) in Ledoit and Péché (2011). Instead of setting $E|z_{ij}|^{12} \leq B$ for some constant B as needed in assumption A1 in Ledoit and Wolf (2012), we have relaxed this, and only need $E|z_{ij}|^{2k} = O(p^{k/2-1})$, $k = 2, 3, 4, 5, 6$ in assumption (A1). It means that we require the 4-th order moments to be bounded asymptotically, but the higher order moments (up to 12th order) can diverge to infinity as $n, p \to \infty$.

2.1 The Marčenko-Pastur law and its generalization

We introduce further notations and definitions in order to present the Marčenko-Pastur law and its generalizations. The Stieltjes transform of a nondecreasing function G is given by

$$m_G(z) = \int_{\mathbb{R}} \frac{1}{\lambda - z} G(\lambda), \quad \text{for any } z \in \mathbb{C}^+. \tag{2.2}$$

Here, $\mathbb{C}^+$ represents the upper half of the complex plane, where the imaginary part of any complex numbers are strictly positive. If G is continuous at a and b, then the following inversion formula for the Stieltjes transform holds:

$$G(b) - G(a) = \lim_{\eta \to 0^+} \frac{1}{\pi} \int_a^b \mathrm{Im}[m_G(\lambda + i\eta)] d\lambda, \tag{2.3}$$

where $\mathrm{Im}(z)$ denotes the imaginary part of a complex number.

Suppose the eigen-decomposition of the sample covariance matrix in (2.1) is $\mathbf{S}_n = \mathbf{P}_n \mathbf{D}_n \mathbf{P}_n^{\mathrm{T}}$, where

$$\mathbf{P}_n = (\mathbf{p}_{n,1}, \ldots, \mathbf{p}_{n,p}), \quad \mathbf{D}_n = \mathrm{diag}(\lambda_{n,1}, \ldots, \lambda_{n,p}), \tag{2.4}$$

so that$\lambda_{n,1} \geq \ldots \geq \lambda_{n,p}$ are the eigenvalues of $\mathbf{S}_n$ with corresponding eigenvectors $\mathbf{p}_{n,1}, \ldots, \mathbf{p}_{n,p}$. The notation $\mathrm{diag}(\cdot)$ denotes a diagonal matrix with the listed entries as the diagonal. We then define the e.d.f. of the sample eigenvalues in (2.4) as $F_p(\lambda) = p^{-1} \sum_{i=1}^p \mathbf{1}_{\{\lambda_{n,i} \leq \lambda\}}$. Using (2.2), the Stieltjes transform of F_p will then be (we suppress the subscript n in $\lambda_{n,i}$ and $\mathbf{p}_{n,i}$ hereafter if no ambiguity arises)

$$m_{F_p}(z) = \frac{1}{p} \sum_{i=1}^p \frac{1}{\lambda_i - z} = p^{-1} \mathrm{tr}[(\mathbf{S}_n - z\mathbf{I}_p)^{-1}], \quad z \in \mathbb{C}^+,$$

where $\mathrm{tr}(\cdot)$ is the trace of a square matrix, and $\mathbf{I}_p$ denotes the identity matrix of size p. It is proved in Marčenko and Pastur (1967) (see Marčenko and Pastur (1967) for the exact assumptions used) that F_p converges almost surely (abbreviated as a.s. hereafter) to some non-random limit F at all points of continuity of F. They also discovered the famous equation, later called the Marčenko-Pastur equation, which can be expressed as

$$m_F(z) = \int_{\mathbb{R}} \frac{1}{\tau[1-c-czm_F(z)]-z} dH(\tau), \quad \text{for any } z \in \mathbb{C}^+. \tag{2.5}$$

For the generalization of (2.5), consider $\Theta_p^g(z) = p^{-1}\mathrm{tr}[(\mathbf{S}_n - z\mathbf{I}_p)^{-1}g(\mathbf{\Sigma}_p)]$, where $g(\cdot)$ is in fact a scalar function on the eigenvalues of a matrix, such that if $\mathbf{\Sigma}_p = \mathbf{V}\mathrm{diag}(\tau_1,\ldots,\tau_p)\mathbf{V}^{\mathrm{T}}$, then

$$g(\mathbf{\Sigma}_p) = \mathbf{V}\mathrm{diag}(g(\tau_1),\ldots,g(\tau_p))\mathbf{V}^{\mathrm{T}}.$$

Hence, $m_{F_p}(z) = \Theta_p^g(z)$ with $g \equiv 1$. In Theorem 2 of Ledoit and Péché (2011), under assumptions (H_1) to (H_4) in their paper, it is proved that $\Theta_p^g(z)$ converges a.s. to $\Theta^g(z)$ for any $z \in \mathbb{C}^+$, where

$$\Theta^g(z) = \int_{\mathbb{R}} \frac{1}{\tau[1-c-czm_F(z)]-z} g(\tau)dH(\tau), \quad z \in \mathbb{C}^+. \tag{2.6}$$

By taking $g = Id$, the identity function, the inverse Stieltjes transform of $\Theta_p^g(z)$, denoted by $\Delta_p(x)$, is given by

$$\Delta_p(x) = \frac{1}{p}\sum_{i=1}^{p} \mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i\mathbf{1}_{\{\lambda_i \leq x\}}, \quad x \in \mathbb{R},$$

on all points of continuity of $\Delta_p(x)$. In Theorem 4 of Ledoit and Péché (2011), using also assumptions (H_1) to (H_4) in their paper, it is proved using (2.6) that $\Delta_p(x)$ converges a.s. to $\Delta(x) = \int_{-\infty}^{x} \delta(\lambda)dF(\lambda)$ for $x \in \mathbb{R}/\{0\}$, the inverse Stieltjes transform of $\Theta^{Id}(z)$. To present the function $\delta(\lambda)$, let $\breve{m}_F(\lambda) = \lim_{z\in\mathbb{C}^+\to\lambda} m_F(z)$ for any $\lambda \in \mathbb{R}/\{0\}$, and $\underline{F}(\lambda) = (1-c)\mathbf{1}_{\{\lambda\geq 0\}} + cF(\lambda)$ when $c > 1$, with $\breve{m}_{\underline{F}}(\lambda) = \lim_{z\in\mathbb{C}^+\to\lambda} m_{\underline{F}}(z)$ for any $\lambda \in \mathbb{R}$. The quantities $\breve{m}_F(\lambda)$ and $\breve{m}_{\underline{F}}(\lambda)$ are shown to exist in Silverstein and Choi (1995). We then have for any $\lambda \in \mathbb{R}$,

$$\delta(\lambda) = \begin{cases} \frac{\lambda}{|1-c-c\lambda\breve{m}_F(\lambda)|^2}, & \text{if } \lambda > 0; \\ \frac{1}{(c-1)\breve{m}_{\underline{F}}(0)}, & \text{if } \lambda = 0 \text{ and } c > 1; \\ 0, & \text{otherwise.} \end{cases} \tag{2.7}$$

This result means that the asymptotic quantity that corresponds to $\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i$ is $\delta(\lambda)$, provided that λ corresponds to λ_i, the ith largest eigenvalue of the sample covariance matrix $\mathbf{S}_n$. It means that we can calculate asymptotically the value of $\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i$ using the sample eigenvalue λ_i according to the nonlinear transformation $\delta(\lambda_i)$, which is an estimable quantity from the data. This is the basis of the nonlinear transformation of sample eigenvalues used in Ledoit and Wolf (2012). We shall come back to this result in section 2.3 and section 2.4.

2.2 Regularization by sample splitting

In Abadir et al. (2014), the data $\mathbf{Y} = (\mathbf{y}_1, \ldots, \mathbf{y}_n)$ is split into two parts, say $\mathbf{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$, where $\mathbf{Y}_1$ has size $p \times m$ and $\mathbf{Y}_2$ has size $p \times (n-m)$. The sample covariance matrix for $\mathbf{Y}_i$ is calculated as $\widetilde{\mathbf{\Sigma}}_i = n_i^{-1}\mathbf{Y}_i\mathbf{Y}_i^{\mathrm{T}}$, $i = 1, 2$, where $n_1 = m$ and $n_2 = n - m$. They propose to estimate the covariance matrix by

$$\check{\mathbf{\Sigma}}_m = \mathbf{P}\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{P}_1)\mathbf{P}^{\mathrm{T}}, \tag{2.8}$$

where $\mathbf{P} = \mathbf{P}_n$ as in (2.4), $\mathrm{diag}(A)$ denotes the diagonal matrix with diagonal entries as in A, and $\mathbf{P}_1$ is an orthogonal matrix such that $\widetilde{\mathbf{\Sigma}}_1 = \mathbf{P}_1\mathbf{D}_1\mathbf{P}_1^{\mathrm{T}}$. Their idea is to use the fact that $\mathbf{P}_1$ and $\widetilde{\mathbf{\Sigma}}_2$ are independent to regularize the eigenvalues. Writing $\mathbf{P}_1 = (\mathbf{p}_{11}, \mathbf{p}_{12}, \ldots, \mathbf{p}_{1p})$, the diagonal values in $\mathbf{P}_1^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{P}_1$ are $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}$, $i = 1, \ldots, p$, and they become the eigenvalues of $\check{\mathbf{\Sigma}}_m$ in (2.8).

On the other hand, in light of (2.7) and the descriptions thereafter, $\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}$ will have the asymptotic nonlinear transformation

$$\delta_1(\lambda) = \begin{cases} \frac{\lambda}{|1-c_1-c_1\lambda\breve{m}_{F_1}(\lambda)|^2}, & \text{if } \lambda > 0; \\ \frac{1}{(c_1-1)\breve{m}_{\underline{F_1}}(0)}, & \text{if } \lambda = 0 \text{ and } c_1 > 1; \\ 0, & \text{otherwise,} \end{cases} \tag{2.9}$$

where $c_1 > 0$ is a constant such that $p/n_1 \to c_1$. The distribution function $F_1(\lambda)$ is the nonrandom limit of $F_{1p}(\lambda) = p^{-1}\sum_{i=1}^{p}\mathbf{1}_{\{\lambda_{1i}\leq\lambda\}}$, with $\lambda_{11} \geq \lambda_{12} \geq \ldots \geq \lambda_{1p}$ being the eigenvalues of $\widetilde{\mathbf{\Sigma}}_1$. The quantities $\breve{m}_{F_1}(\lambda)$ and $\breve{m}_{\underline{F_1}}(0)$ are defined in parallel to those in (2.7). We show in Theorem 1 that the quantities $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}$ and $\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}$ are in fact asymptotically the same. Hence, they both correspond to $\delta_1(\lambda_{1i})$, with $\delta_1(\lambda)$ defined in (2.9). This forms the basis for the covariance matrix estimator in our paper, and will be explained in full details in section 2.3.

Remark 1. If $n_1/n \to 1$, then $p/n_1, p/n$ both go to the same limit $c_1 = c > 0$. Theorem 4.1 of Bai and Silverstein (2010) tells us then both F_p and F_{1p} converges to the same limit almost surely under the assumptions (A1) to (A4). That is, $F = F_1$ almost surely, and hence $\delta_1(\cdot) = \delta(\cdot)$. This implies that $\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i$ and $\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}$ are asymptotically almost surely the same.

2.3 The covariance matrix estimator

In this paper, we use the sample splitting idea in section 2.2 and split the data into $\mathbf{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$, so that $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are independent of each other by our assumption of independence. We calculate $\widetilde{\mathbf{\Sigma}}_i = n_i^{-1}\mathbf{Y}_i\mathbf{Y}_i^{\mathrm{T}}$ with $n_1 = m$ and $n_2 = n - m$, and carry out the eigen-decomposition $\widetilde{\mathbf{\Sigma}}_1 = \mathbf{P}_1\mathbf{D}_1\mathbf{P}_1^{\mathrm{T}}$ as in section 2.2.

Using $\|A\|_F = \text{tr}^{1/2}(AA^{\mathrm{T}})$ to denote the Frobenius norm of a matrix A, we then consider the following optimization problem:

$$\min_{\mathbf{D}} \|\mathbf{P}_1\mathbf{D}\mathbf{P}_1^{\mathrm{T}} - \mathbf{\Sigma}_p\|_F, \text{ where } \mathbf{D} \text{ is a diagonal matrix.} \tag{2.10}$$

Essentially, we are considering the class of rotation-equivariant estimator $\widehat{\mathbf{\Sigma}}_m = \mathbf{P}_1\mathbf{D}\mathbf{P}_1^{\mathrm{T}}$, where m is the location we split the data matrix $\mathbf{Y}$ into two. Basic calculus shows that the optimum is achieved at $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$, where

$$d_i = \mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}, \quad i = 1, \dots, p. \tag{2.11}$$

This is however unknown to us since $\mathbf{\Sigma}_p$ is unknown. We can see now that why we are interested in the asymptotic properties of $\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}$, which corresponds asymptotically to $\delta_1(\lambda_{1i})$ as defined in (2.9). The important thing about $\delta_1(\lambda_{1i})$ is that this function depends on the sample eigenvalues λ_{1i} of $\widetilde{\mathbf{\Sigma}}_1$, which are immediately available, and other quantities inside the definition of $\delta_1(\lambda_{1i})$ are also estimable from data.

Instead of estimating $\breve{m}_{F_1}(\lambda_{1i})$ and $\breve{m}_{\underline{F_1}}(0)$ as contained in the expression in $\delta_1(\lambda_{1i})$ in (2.9), which is basically what the paper Ledoit and Wolf (2012) is about (they estimate $\breve{m}_F(\lambda_i)$ and $\breve{m}_{\underline{F}}(0)$ in $\delta(\lambda_i)$ in (2.7) in their paper, that is, without splitting the data at all), we consider the asymptotic properties of $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}$. To this end, define the function

$$\Psi_m^{(1)}(z) = \frac{1}{p}\text{tr}[(\widetilde{\mathbf{\Sigma}}_1 - z\mathbf{I}_p)^{-1}\widetilde{\mathbf{\Sigma}}_2], \quad z \in \mathbb{C}^+. \tag{2.12}$$

We can show that the inverse Stieltjes transform of this function is, on all points of continuity of $\Phi_m^{(1)}$,

$$\Phi_m^{(1)}(x) = \frac{1}{p}\sum_{i=1}^{p} \mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}\mathbf{1}_{\{\lambda_{1i}\leq x\}}, \quad x \in \mathbb{R}. \tag{2.13}$$

Similar to section 2.1, we can carry out asymptotic analysis on $\Phi_m^{(1)}(x)$ in order to study the asymptotic behavior of $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}$ for each i. The results are shown in the following theorem.

Theorem 1. *Let assumptions (A1) to (A4) be satisfied. Suppose $p/n_1 \to c_1 > 0$ and $c_1 \neq 1$. Assume also $\sum_{n\geq 1} n_2^{-3} < \infty$. We have the following:*

(i) The function $\Psi_m^{(1)}(z)$ defined in (2.12) converges a.s. to a non-random limit $\Psi^{(1)}(z)$ for any $z \in \mathbb{C}^+$, defined by

$$\Psi^{(1)}(z) = \frac{1}{c_1(1 - c_1 - c_1 z m_{F_1}(z))} - \frac{1}{c_1},$$

where F_1 and c_1 are defined in equation (2.9) and the descriptions therein.

(ii) The inverse Stieltjes transform of $\Psi^{(1)}(z)$ is $\Phi^{(1)}(x) = \int_{-\infty}^{x} \delta_1(\lambda) dF_1(\lambda)$ on all points of continuity of $\Phi^{(1)}(x)$, where $\delta_1(\lambda)$ is given in (2.9).

(iii) The function $\Phi_m^{(1)}(x)$ defined in (2.13) converges a.s. to $\Phi^{(1)}(x)$ on all points of continuity of $\Phi^{(1)}(x)$.

Moreover, if $c_1 = 1$, we still have $\mathbf{\Phi}_m^{(1)}(x) - p^{-1}\sum_{i=1}^{p} \mathbf{p}_{1i}^{\mathrm{T}} \mathbf{\Sigma}_p \mathbf{p}_{1i} \mathbf{1}_{\{\lambda_{1i} \leq x\}}$ converges a.s. to 0 as $n_1, p \to \infty$, so that $p/n_1 \to 1$ and $\sum_{n \geq 1} n_2^{-3} < \infty$.

Part (iii) of this theorem shows that the asymptotic nonlinear transformation for $\mathbf{p}_{1i}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}_2 \mathbf{p}_{1i}$ is given by (2.9), which is the same asymptotic nonlinear transformation for $d_i = \mathbf{p}_{1i}^{\mathrm{T}} \mathbf{\Sigma}_p \mathbf{p}_{1i}$ in (2.11). This means that $\mathbf{p}_{1i}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}_2 \mathbf{p}_{1i}$ and $\mathbf{p}_{1i}^{\mathrm{T}} \mathbf{\Sigma}_p \mathbf{p}_{1i}$ are asymptotically the same. Note that this conclusion is true even when $c_1 = 1$ by the very last part of the theorem, which is a case excluded in Ledoit and Wolf (2012) and Ledoit and Wolf (2013a). With this, we propose our covariance matrix estimator as

$$\widehat{\mathbf{\Sigma}}_m = \mathbf{P}_1 \mathrm{diag}(\mathbf{P}_1^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}_2 \mathbf{P}_1) \mathbf{P}_1^{\mathrm{T}}. \tag{2.14}$$

This is almost the same as $\check{\mathbf{\Sigma}}_m$ in (2.8), except that $\mathbf{P}$ there is replaced by $\mathbf{P}_1$. This makes sense with respect to minimizing the Frobenius loss in (2.10), since Theorem 1 shows that $\mathbf{p}_{1i}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}_2 \mathbf{p}_{1i}$ is asymptotically the same as the minimizer d_i in (2.11).

It appears that the estimator $\widehat{\mathbf{\Sigma}}_m$ is not using as much information as $\check{\mathbf{\Sigma}}_m$, since the eigenmatrix $\mathbf{P}_1$ uses only information on $\mathbf{Y}_1$ but not the full set of data $\mathbf{Y}$. However, coupled with averaging to be introduced in section 4.1, simulation experiments in section 5 shows that our estimator can have comparable or even better performance than $\check{\mathbf{\Sigma}}_m$ or its averaging counterparts introduced in Abadir et al. (2014). A heuristic argument is introduced in section 4.1 on why averaging produces a better estimator, and Figure 8 in section 5 shows explicitly why averaging using a slightly smaller data set $\mathbf{Y}_1$ is better than not averaging while using the full set of data $\mathbf{Y}$. We will come back to this point in the descriptions following Figure 8.

While constructing $\widehat{\mathbf{\Sigma}}_m$ involves only splitting the data into two portions and carrying out eigen-analysis for one of them, the estimator proposed in Ledoit and Wolf (2012) requires the estimation of $\breve{m}_F(\lambda_i)$ for each $i = 1, \ldots, p$, which can be computationally expensive. By inspecting the form of the nonlinear transformation in (2.7), when $c > 1$, the term $\breve{m}_{\underline{F}}(0)$ has to be estimated as well, which requires special attention, and is not dealt with in Ledoit and Wolf (2012), although Ledoit and Wolf (2013b) has addressed this and extended their package to deal with $c > 1$. The estimator $\widehat{\mathbf{\Sigma}}_m$, on the other hand, is calculated the same way no matter $c_1 < 1$ or $c_1 \geq 1$. In section 4, we propose an improvement to $\widehat{\mathbf{\Sigma}}_m$ through averaging, and compare the performance and speed of calculating this improved version of $\widehat{\mathbf{\Sigma}}_m$

with the one in Ledoit and Wolf (2013b). Comparisons will also be carried out with the grand average estimator proposed in equation (18) of Abadir et al. (2014).

Intuitively, the choice of m is important for the performance of the estimator, and it seems that we should use as much data to estimate $\widetilde{\mathbf{\Sigma}}_1$ as possible since $\mathbf{P}_1$ plays an important role. However, in Theorem 5, the sample size n_2 for constructing $\widetilde{\mathbf{\Sigma}}_2$ has to go to infinity with n, albeit at a slower rate, in order for our estimator to be asymptotically efficient. We also demonstrate empirically in section 5 that n_2 has to be reasonably large for the estimator to perform well in practice.

2.4 The precision matrix estimator

We use the inverse of $\widehat{\mathbf{\Sigma}}_m$ in (2.14) as our precision matrix estimator, that is,

$$\widehat{\mathbf{\Omega}}_m = \widehat{\mathbf{\Sigma}}_m^{-1}. \tag{2.15}$$

With respect to the inverse Stein's loss function SL for estimating $\mathbf{\Omega}_p = \mathbf{\Sigma}_p^{-1}$, where

$$SL(\mathbf{\Omega}_p, \widehat{\mathbf{\Omega}}) = \mathrm{tr}(\mathbf{\Omega}_p^{-1}\widehat{\mathbf{\Omega}}) - \log|\mathbf{\Omega}_p^{-1}\widehat{\mathbf{\Omega}}| - p, \tag{2.16}$$

an asymptotically optimal estimator is indeed $\widehat{\mathbf{\Omega}}_m = \widehat{\mathbf{\Sigma}}_m^{-1}$ given in (2.15), as shown in Proposition 2 below. This inverse Stein's loss function is also introduced in Ledoit and Wolf (2013a). In Theorem 4.1 of their paper, they proved that the nonlinear transformation depicted in equation (2.7) of our paper is in fact optimal with respect to asymptotically minimizing this loss function.

The Stein's loss function first appeared in James and Stein (1961) for measuring the error of estimating $\mathbf{\Sigma}_p$ by $\widehat{\mathbf{\Sigma}}$. The inverse Stein's loss function is also a scaled version of the Kullback-Leibler divergence of the normal distribution $N(\mathbf{0}, \mathbf{\Sigma}_p)$ relative to $N(\mathbf{0}, \widehat{\mathbf{\Sigma}})$. We provide an alternative formulation from the results of Theorem 4.1 of Ledoit and Wolf (2013a), showing that the precision matrix estimator $\widehat{\mathbf{\Omega}}_m$ in equation (2.15) does indeed minimize the loss in (2.16) asymptotically.

Proposition 2. *Consider the class of estimators $\widehat{\mathbf{\Omega}} = \mathbf{P}_1\mathbf{D}_1^{-1}\mathbf{P}_1^{\mathrm{T}}$, where $\mathbf{D}_1^{-1} = diag(d_1^{-1}, \ldots, d_p^{-1})$. Then the optimization problem*

$$\min_{d_i} SL(\mathbf{\Omega}_p, \widehat{\mathbf{\Omega}})$$

has a solution given by $d_i = \mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}$, $i = 1, \ldots, p$.

Since Theorem 1 (iii) shows that the asymptotic nonlinear transformation for $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}$ is the same as that for $d_i = \mathbf{p}_{1i}^{\mathrm{T}} \boldsymbol{\Sigma}_p \mathbf{p}_{1i}$, the above proposition immediately implies that

$$\widehat{\boldsymbol{\Omega}}_m = \mathbf{P}_1[\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}} \widetilde{\boldsymbol{\Sigma}}_2 \mathbf{P}_1)]^{-1} \mathbf{P}_1^{\mathrm{T}} = \widehat{\boldsymbol{\Sigma}}_m^{-1}$$

is an asymptotically optimal estimator for $\boldsymbol{\Omega}_p$.

Proof of Proposition 2. With $\widehat{\boldsymbol{\Omega}} = \mathbf{P}_1 \mathbf{D}_1^{-1} \mathbf{P}_1^{\mathrm{T}}$ where $\mathbf{P}_1 = (\mathbf{p}_{11}, \ldots, \mathbf{p}_{1p})$, we have

$$\begin{aligned} SL(\boldsymbol{\Omega}_p, \widehat{\boldsymbol{\Omega}}) &= \mathrm{tr}(\boldsymbol{\Sigma}_p \mathbf{P}_1 \mathbf{D}_1^{-1} \mathbf{P}_1^{\mathrm{T}}) - \log |\boldsymbol{\Sigma}_p \mathbf{P}_1 \mathbf{D}_1^{-1} \mathbf{P}_1^{\mathrm{T}}| - p \\ &= \sum_{i=1}^{p} \frac{\mathbf{p}_{1i}^{\mathrm{T}} \boldsymbol{\Sigma}_p \mathbf{p}_{1i}}{d_i} - \log |\mathbf{D}_1^{-1}| - \log |\mathbf{P}_1 \mathbf{P}_1^{\mathrm{T}}| - \log |\boldsymbol{\Sigma}_p| - p \\ &= \sum_{i=1}^{p} \left(\frac{\mathbf{p}_{1i}^{\mathrm{T}} \boldsymbol{\Sigma}_p \mathbf{p}_{1i}}{d_i} + \log(d_i) \right) - (p + \log |\boldsymbol{\Sigma}_p|), \end{aligned}$$

which is clearly minimized at $d_i = \mathbf{p}_{1i}^{\mathrm{T}} \boldsymbol{\Sigma}_p \mathbf{p}_{1i}$ for $i = 1, \ldots, p$. $\square$

In section 4, we propose an improvement on the estimator $\widehat{\boldsymbol{\Sigma}}_m$, and we use its inverse as an improvement to the precision matrix estimator $\widehat{\boldsymbol{\Omega}}_m$.

3 Extension to Data from a Factor Model

Results from previous sections rely heavily on assumption (A1) in section 2, that $\mathbf{y}_i = \boldsymbol{\Sigma}_p^{1/2} \mathbf{z}_i$ with $\mathbf{z}_i$ having p independent and identically distributed components, and that p goes to infinity together with n. This is also an assumption on which the results from Ledoit and Wolf (2012) rely. However, even for a random sample of p-dimensional vectors, this may not always be true. For instance, consider the factor model defined by

$$\mathbf{y}_i = \mathbf{A}\mathbf{x}_i + \boldsymbol{\epsilon}_i, \quad i = 1, \ldots, n, \tag{3.17}$$

with $\mathbf{A}$ being a $p \times r$ factor loading matrix, $\mathbf{x}_i$ an $r \times 1$ vector of factors and $\boldsymbol{\epsilon}_i$ a $p \times 1$ vector of noise series. We assume that r is much smaller than p, so that a large number of p components in $\mathbf{y}_i$ have dynamics driven by the small number of r factors. In Lam et al. (2011), they consider $\{\mathbf{x}_i\}$ being a stationary time series and $\{\boldsymbol{\epsilon}_i\}$ being a white noise. In this paper, we assume that $\{\mathbf{x}_i\}$ has independent and identically distributed vectors, and the same goes for $\{\boldsymbol{\epsilon}_i\}$. Hence, the $\mathbf{y}_i$'s are independent and identically distributed. A factor model is often used in modeling a panel of stock returns in finance. See Fan et al. (2008), for example. Also, see the data analysis of the return series of S&P500 constituents in Lam and Yao (2012). There, two factors are found, and 97.7% of the variation of the return series can be explained by a linear combination of the two factors.

The covariance matrix for $\mathbf{y}_i$ in model (3.17) can be written as

$$\boldsymbol{\Sigma}_p = \mathbf{A}\boldsymbol{\Sigma}_x\mathbf{A}^{\mathrm{T}} + \boldsymbol{\Sigma}_\epsilon, \tag{3.18}$$

where $\boldsymbol{\Sigma}_x = \mathrm{var}(\mathbf{x}_i)$ and $\boldsymbol{\Sigma}_\epsilon = \mathrm{var}(\boldsymbol{\epsilon}_i)$. We can easily see that with the low dimensionality of $\mathbf{x}_i$, if some of the factors in $\mathbf{x}_i$ are strong factors (that is, the corresponding columns in $\mathbf{A}$ have the majority of coefficients being non-zero; see Lam et al. (2011) and Lam and Yao (2012) for the formal definitions of strong and weak factors), we cannot really write $\mathbf{y}_i = \boldsymbol{\Sigma}_p^{1/2}\mathbf{y}_i^*$ with $\mathbf{y}_i^*$ having p independent components, since some factors in $\mathbf{x}_i$ are shared in most of the components of $\mathbf{y}_i$. Hence, assumption (A1) in section 2 cannot be satisfied, and the method in Ledoit and Wolf (2012) cannot be asymptotically optimal. We show in Theorem 3 below that our estimator $\widehat{\boldsymbol{\Sigma}}_m$ in (2.14), on the other hand, is still asymptotically optimal in the same sense as in sections 2.3 and 2.4.

Before presenting the main result of this section, we present the following assumptions for the factor model in (3.17). They are parallel to the assumptions (A1) to (A2) in section 2.

(F1) The series $\{\boldsymbol{\epsilon}_i\}$ has $\boldsymbol{\epsilon}_i = \boldsymbol{\Sigma}_\epsilon^{1/2}\boldsymbol{\xi}_i$, where $\boldsymbol{\xi}_i$ is a $p \times 1$ vector of independent and identically distributed random variables ξ_{ij}. Each ξ_{ij} has mean 0 and unit variance, and $E|\xi_{ij}|^{2k} = O(p^{k/2-1})$ for $k = 2, 3, 4, 5, 6$. The factor series $\{\mathbf{x}_t\}$ has a constant dimension r, and $\mathbf{x}_t = \boldsymbol{\Sigma}_x^{1/2}\mathbf{x}_t^*$ where $\mathbf{x}_t^*$ is a $r \times 1$ vector of independent and identically distributed random variables x_{ti}^*. Also, $E|x_{tj}^*|^{2k} < \infty$ for each t, j for $k = 2, 3, 4, 5, 6$.

(F2) The covariance matrix $\boldsymbol{\Sigma}_x = \mathrm{var}(\mathbf{x}_i)$ is such that $\|\boldsymbol{\Sigma}_x\| = O(1)$. The covariance matrix $\boldsymbol{\Sigma}_\epsilon = \mathrm{var}(\boldsymbol{\epsilon}_i)$ also has $\|\boldsymbol{\Sigma}_\epsilon\| = O(1)$. Both covariance matrices are non-random. The factor loading matrix $\mathbf{A}$ is such that $\|\mathbf{A}\|_F^2 = \mathrm{tr}(\mathbf{A}\mathbf{A}^{\mathrm{T}}) = O(p)$.

The assumption $\|\mathbf{A}\|_F^2 = O(p)$ entails both strong and weak factors as defined in Lam et al. (2011).

Theorem 3. *Let assumptions (F1) and (F2) be satisfied. Assume that we split the data like that in section 2.3, with p and n_2 both go to infinity together and $p/n_1 \to c_1 > 0$. Assume also $\sum_{n\geq 1} n_2^{-3} < \infty$. Then with $\boldsymbol{\Sigma}_p$ the covariance matrix of $\mathbf{y}_i$ as defined in (3.18), we have for almost all $x \in \mathbb{R}$,*

$$\frac{1}{p}\sum_{i=1}^{p}\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i}\mathbf{1}_{\{\lambda_{1i}\leq x\}} - \frac{1}{p}\sum_{i=1}^{p}\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}\mathbf{1}_{\{\lambda_{1i}\leq x\}} \xrightarrow{a.s.} 0.$$

Like Theorem 1, this theorem says that $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i}$ is asymptotically equal to $\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}$ for each $i = 1, \ldots, p$. Since $d_i = \mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}$ (see (2.11)) is the optimal solution for the optimization problem (2.10), it means that the covariance matrix estimator $\widehat{\boldsymbol{\Sigma}}_m$ in (2.14), under the factor model setting in this section, is still asymptotical optimal for estimating $\boldsymbol{\Sigma}_p$ in (3.18) with respect to the Frobenius loss, when considering the class of rotation-equivariant estimators $\widehat{\boldsymbol{\Sigma}}(\mathbf{D}) = \mathbf{P}_1\mathbf{D}\mathbf{P}_1^{\mathrm{T}}$. At the same time, by Proposition 2, the inverse estimator $\widehat{\boldsymbol{\Omega}}_m = \widehat{\boldsymbol{\Sigma}}_m^{-1}$ is also an asymptotically optimal estimator of the precision matrix $\boldsymbol{\Omega}_p = \boldsymbol{\Sigma}_p^{-1}$ with respect to the inverse Stein's loss function (2.16).

The optimality result in Theorem 3 means that we do not need to know the exact asymptotic nonlinear transformation to which $\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}$ converges in order to find an optimal covariance or precision matrix estimator. On the other hand, the form of the nonlinear transformation is crucial for the method in Ledoit and Wolf (2012) to work. In this sense, our estimators are more robust to changes in the structure of the data. We demonstrate the performance of the estimator when the data follows a factor model in section 5.

4 Asymptotic Efficiency Loss and Practical Implementation

In this section, we introduce the following ideal estimator,

$$\widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal}} = \mathbf{P}\mathrm{diag}(\mathbf{P}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{P})\mathbf{P}^{\mathrm{T}}. \tag{4.1}$$

This is also called the finite-sample optimal estimator in Ledoit and Wolf (2012). Compare to $\widehat{\boldsymbol{\Sigma}}_m$ in (2.14), this ideal estimator used the full set of data for calculating the eigenmatrix $\mathbf{P}$, and it assumed the knowledge of $\boldsymbol{\Sigma}_p$ itself instead of using $\widetilde{\boldsymbol{\Sigma}}_2$ to estimate it like our estimator does. With this ideal estimator, we define the efficiency loss of an estimator $\widehat{\boldsymbol{\Sigma}}$ as

$$EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}) = 1 - \frac{L(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal}})}{L(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}})}, \tag{4.2}$$

where $L(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}})$ is a loss function for estimating $\boldsymbol{\Sigma}_p$ by $\widehat{\boldsymbol{\Sigma}}$. The two loss functions we focus on in this paper are the Frobenius loss

$$L(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}) = \left\|\widehat{\boldsymbol{\Sigma}} - \boldsymbol{\Sigma}_p\right\|_F^2, \tag{4.3}$$

and the inverse Stein's loss introduced in (2.16), which, in terms of $\widehat{\boldsymbol{\Sigma}}$ and $\boldsymbol{\Sigma}_p$, is

$$L(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}) = \mathrm{tr}\left(\boldsymbol{\Sigma}_p\widehat{\boldsymbol{\Sigma}}^{-1}\right) - \log\det\left(\boldsymbol{\Sigma}_p\widehat{\boldsymbol{\Sigma}}^{-1}\right) - p. \tag{4.4}$$

If $EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}) \leq 0$, it means that the estimator $\widehat{\boldsymbol{\Sigma}}$ is doing at least as good as the ideal estimator $\widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal}}$ in terms of the loss function L, and vice versa. To present the asymptotic efficiency results with respect to these two loss functions, we need to assume the following set of assumptions (A1)' and (A2)':

(A1)' Each observation can be written as $\mathbf{y}_i = \mathbf{\Sigma}_p^{1/2}\mathbf{z}_i$ for $i = 1, \ldots, n$, where each $\mathbf{z}_i$ is a $p \times 1$ vector of independent and identically distributed random variables z_{ij}. Each z_{ij} has mean 0 and unit variance, and $E|z_{ij}|^k \leq B < \infty$ for some constant B and $2 < k \leq 20$.

(A2)' The population covariance matrix is non-random and of size $p \times p$. Furthermore, $\|\mathbf{\Sigma}_p\| = O(1)$.

Or, if the data follows a factor model $\mathbf{y}_i = \mathbf{A}\mathbf{x}_i + \boldsymbol{\epsilon}_i$, we need to assume the following set of assumptions (F1)' and (F2)':

(F1)' The series $\{\boldsymbol{\epsilon}_i\}$ has $\boldsymbol{\epsilon}_i = \mathbf{\Sigma}_\epsilon^{1/2}\boldsymbol{\xi}_i$, where $\boldsymbol{\xi}_i$ is a $p \times 1$ vector of independent and identically distributed random variables ξ_{ij}. Each ξ_{ij} has mean 0 and unit variance, and $E|\xi_{ij}|^k \leq B < \infty$ for some constant B and $k \leq 20$. The factor series $\{\mathbf{x}_t\}$ has a constant dimension r, and $\mathbf{x}_t = \mathbf{\Sigma}_x^{1/2}\mathbf{x}_t^*$ where $\mathbf{x}_t^*$ is a $r \times 1$ vector of independent and identically distributed random variables x_{ti}^*. Also, $E|x_{tj}^*|^k \leq B < \infty$ for some constant B and $2 < k \leq 20$.

(F2)' Same as (F2), meaning that $\|\mathbf{\Sigma}_x\|, \|\mathbf{\Sigma}_\epsilon\| = O(1)$ and $\|\mathbf{A}\|_F^2 = O(p)$.

Assumptions (A1)' and (A2)' are parallel to (A1) and (A2) respectively. The more restrictive moments assumptions are needed for the proof of Lemma 1, which is important for proving Corollary 4 and the asymptotic efficiency results in Theorem 5. Assumption (F1)' is parallel to (F1), and is for data with a factor structure.

Lemma 1. *Let assumption (A1)' be satisfied. If the split location m is such that $\sum_{n\geq 1} p(n-m)^{-5} < \infty$, we have*

$$\max_{1\leq i\leq p}\left|\frac{\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i} - \mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}}{\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}}\right| \xrightarrow{a.s.} 0.$$

The same holds true if the data is from a factor model, with assumption (F1)' satisfied together with $\sum_{n\geq 1} p(n-m)^{-5} < \infty$.

This convergence of $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}$ for all i is stronger than those in Theorems 1 and 3. Hence, more restrictive assumptions are needed. The proof of this Lemma is in the Appendix. With the result in Lemma 1, it is easy to see the following.

Corollary 4. *Let the assumptions in Lemma 1 hold. Then as $n, p \to \infty$ almost surely, $\widehat{\mathbf{\Sigma}}_m$ is positive definite as long as $\mathbf{\Sigma}_p$ is.*

Proof of Corollary 4. Note that $\widehat{\mathbf{\Sigma}}_m$ is always positive semi-definite by construction, since all the eigenvalues $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}$, $i = 1, \ldots, p$, are non-negative. The convergence result in Lemma 1 ensures that all the eigenvalues of $\widehat{\mathbf{\Sigma}}_m$ are almost surely larger than $(1-\epsilon)\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i} \geq (1-\epsilon)\lambda_{\min}(\mathbf{\Sigma}_p) > 0$ for large enough n and a fixed $0 < \epsilon < 1$, if $\mathbf{\Sigma}_p$ is positive definite. $\square$

This result is not formally proved in Ledoit and Wolf (2012, 2013a,b), and in fact the conclusion can be wrong for their nonlinear shrinkage estimator when the data follows a factor model. Our simulation results in section 5 do show that the nonlinear shrinkage estimator is singular in some simulation runs for data following a factor model, when our estimator is still positive definite.

Corollary 4 is also different from Proposition 1 in Abadir et al. (2014). For their proof to be valid, they in fact need p to be smaller than $n - m$ (in our notations), otherwise the estimator is only at most positive-semi definite. Our Corollary 4, on the other hand, allows for p to be even larger than n.

With Lemma 1, we can also prove that $\widehat{\mathbf{\Sigma}}_m$ is asymptotically efficient relative to $\widehat{\mathbf{\Sigma}}_{\text{Ideal}}$ with respect to both the Frobenius and the inverse Stein's losses, as long as the split location m satisfies some conditions.

Theorem 5. *Let assumptions (A1)', (A2)', (A3) and (A4) be satisfied. Assume the split location m for $\widehat{\mathbf{\Sigma}}_m$ is such that $m/n \to 1$ and $n - m \to \infty$ as $n \to \infty$, with $\sum_{n\geq 1} p(n-m)^{-5} < \infty$. We then have $EL(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_m) \overset{a.s.}{\to} 0$ with respect to both the Frobenius and the inverse Stein's loss functions, as long as $\mathbf{\Sigma}_p \neq \sigma^2\mathbf{I}$.*

The proof is relegated to the Appendix. Note that we exclude the case $\mathbf{\Sigma}_p = \sigma^2\mathbf{I}_p$, which has zero loss for the ideal estimator. In section 5, we include this case to compare the performance of various methods including our estimator.

The results from the above theorem show that $\widehat{\mathbf{\Sigma}}_m$ is asymptotically the same as the ideal estimator $\widehat{\mathbf{\Sigma}}_{\text{Ideal}}$ with respect to the Frobenius or inverse Stein's losses. Since $p/n \to c > 0$, a choice of m that satisfies all the conditions is $m = n - an^{1/2}$ for some constant $a > 0$. In practice, this form of split location works well, and we provide a way in section 4.2 to identify an m for good performance.

In Theorem 5, data from a factor model is excluded. We do not pursue the proof here, although we still conjecture that $EL(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_m) \overset{a.s.}{\to} 0$ for the inverse Stein's loss. See the empirical results in section 5 for details.

4.1 Improvement with averaging

We can improve the performance of $\widehat{\mathbf{\Sigma}}_m$ in (2.14) by noting that each vector $\mathbf{y}_i$ in $\mathbf{Y}$ is independent of each other and identically distributed. We can permute the data, form another data matrix $\mathbf{Y}^{(j)}$, and split the data into two independent parts $\mathbf{Y}^{(j)} = (\mathbf{Y}_1^{(j)}, \mathbf{Y}_2^{(j)})$ as in section 2.3. Then we can form another estimator

$$\widehat{\mathbf{\Sigma}}_m^{(j)} = \mathbf{P}_{1j}\mathrm{diag}(\mathbf{P}_{1j}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(j)}\mathbf{P}_{1j})\mathbf{P}_{1j}^{\mathrm{T}}, \text{ where } \widetilde{\mathbf{\Sigma}}_i^{(j)} = n_i^{-1}\mathbf{Y}_i^{(j)}\mathbf{Y}_i^{(j)\mathrm{T}} \text{ with } m = n_1,\ n = n_1 + n_2, \tag{4.5}$$

for $j = 1, \ldots, M$. Each j represents a permutation of the data so that no two $\mathbf{Y}_1^{(j)}$'s contain exactly the same data, thus $M \leq \binom{n}{m}$. The matrix $\mathbf{P}_{1j}$ contains the orthonormal set of eigenvectors such that $\widetilde{\mathbf{\Sigma}}_1^{(j)} = \mathbf{P}_{1j}\mathbf{D}_{1j}\mathbf{P}_{1j}^{\mathrm{T}}$.

In Abadir et al. (2014), they improve the performance of their estimator by averaging the regularized eigenvalues over different j and different split location m. However, we know from Theorem 1 and Proposition 2 that the regularized eigenvalues $\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}$ are asymptotically optimal only when coupled with $\mathbf{P}_1$ to form the estimator $\widehat{\mathbf{\Sigma}}_m$. Hence, for each j, the regularized eigenvalues in $\mathrm{diag}(\mathbf{P}_{1j}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(j)}\mathbf{P}_{1j})$ are asymptotically optimal only when coupled with $\mathbf{P}_{1j}$ to calculate $\widehat{\mathbf{\Sigma}}_m^{(j)}$, as in (4.5). This forbid us from averaging the eigenvalues in $\mathrm{diag}(\mathbf{P}_{1j}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(j)}\mathbf{P}_{1j})$ over the index j, since each set is only asymptotically optimal when coupled with $\mathbf{P}_{1j}$, and it can be suboptimal to couple the averaged set of eigenvalues with other orthogonal matrix $\mathbf{P}$.

In light of the above argument, we average the sum of $\widehat{\mathbf{\Sigma}}_m^{(j)}$ to improve the performance by setting

$$\widehat{\mathbf{\Sigma}}_{m,M} = \frac{1}{M}\sum_{j=1}^{M}\widehat{\mathbf{\Sigma}}_m^{(j)}, \tag{4.6}$$

where $\widehat{\mathbf{\Sigma}}_m^{(j)}$ is as given in (4.5). This is different from equation (15) in Abadir et al. (2014) as we have argued in the paragraph before. However, they have proved in Proposition 3 in their paper that the expected elementwise loss in L_1 and L_2 norm for their estimator is asymptotically optimized if the split location m is such that $m, n - m \to \infty$ with $m/n \to \gamma \in (0, 1)$. This choice of split is certainly excluded from our results in Theorem 5, where we need $m/n \to 1$. The reason for this major difference is that $p \to \infty$ as $n \to \infty$ such that $p/n \to c > 0$ in our paper, whereas p is treated as fixed in Abadir et al. (2014) (except for their proposition 5, where p can diverge to infinity, but still need to be at a rate slower than n). In our simulations in section 5, we demonstrate that the split m with $m/n \to \gamma \in (0, 1)$ can be suboptimal if p is indeed growing with n.

We provide an efficiency result for the averaged estimator $\widehat{\mathbf{\Sigma}}_{m,M}$ below.

Theorem 6. *Let the assumptions in Theorem 5 be satisfied. Assume further that M is finite, and $\tau_{n,p}$, the smallest eigenvalue of $\mathbf{\Sigma}_p$, is bounded uniformly away from 0. We then have $EL(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_{m,M}) \leq 0$ almost surely with respect to the Frobenius loss or the inverse Stein's loss function as $n, p \to \infty$, as long as $\mathbf{\Sigma}_p \neq \sigma^2\mathbf{I}$.*

This theorem shows that the estimator $\widehat{\mathbf{\Sigma}}_{m,M}$ also enjoys asymptotic efficiency with respect to the ideal estimator. The proof is relegated to the Appendix.

We demonstrate heuristically why $\widehat{\mathbf{\Sigma}}_{m,M}$ in (4.6) is better than $\widehat{\mathbf{\Sigma}}_m$. It is not difficult to show that the Frobenius losses for $\widehat{\mathbf{\Sigma}}_m$ and $\widehat{\mathbf{\Sigma}}_{m,M}$ have

$$\begin{aligned} p^{-1}\big\|\widehat{\mathbf{\Sigma}}_m - \mathbf{\Sigma}_p\big\|_F^2 &= p^{-1}\sum_{i=1}^{p}(\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i} - \mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i})^2 + p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m} - \mathbf{\Sigma}_p\big\|_F^2, \\ p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{m,M} - \mathbf{\Sigma}_p\big\|_F^2 &\leq \frac{2}{M}\sum_{s=1}^{M}p^{-1}\sum_{i=1}^{p}(\mathbf{p}_{1s,i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(s)}\mathbf{p}_{1s,i} - \mathbf{p}_{1s,i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1s,i})^2 + \frac{2}{M}\sum_{s=1}^{M}p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m}^{(s)} - \mathbf{\Sigma}_p\big\|_F^2, \end{aligned} \tag{4.7}$$

where $\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m}^{(s)} = \mathbf{P}_{1s}\mathrm{diag}(\mathbf{P}_{1s}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}_{1s})\mathbf{P}_{1s}^{\mathrm{T}}$, and $\mathbf{p}_{1s,i}$ is the ith column of $\mathbf{P}_{1s}$. Assuming conditions (A1)' and (A2)' here for instance, then the first terms on the right hand side for both losses above have similar order of upper bound for their variances, and they are both going to 0 almost surely using Lemma 1. Since $p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m} - \mathbf{\Sigma}_p\big\|_F^2$ is not going to 0 almost surely by our assumption, both second terms on the right hand side of (4.7) dominate their respective first terms. However, the variance for $\frac{2}{M}\sum_{i=1}^{M}p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m}^{(s)} - \mathbf{\Sigma}_p\big\|_F^2$ can be much smaller than $p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m} - \mathbf{\Sigma}_p\big\|_F^2$. This is because the correlation between two $\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m}^{(s)}$ for different s can be small, as they depend only on the first part of the split data, which can be different after permutation. Hence, the variance for $p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{m,N} - \mathbf{\Sigma}_p\big\|_F^2$ is usually smaller than that for $p^{-1}\big\|\widehat{\mathbf{\Sigma}}_m - \mathbf{\Sigma}_p\big\|_F^2$.

A larger M usually gives better performance, but becomes computationally expensive when it is too large. Luckily, our simulation results in section 5 demonstrate that $M = 50$ attains a very good performance already in a variety of settings, when a random permutation of the data is used.

4.2 Choice of split location

Rather than averaging over different split locations m like the grand average estimator (15) of Abadir et al. (2014), Theorem 5 suggests that $m = n - an^{1/2}$ can be a choice to achieve asymptotic efficiency when $p/n \to c > 0$, but not so when m/n goes to a constant smaller than 1. Indeed, we find that our estimator does not perform well when m is small, and when m is too close to n, the performance suffers also, which is also demonstrated in section 5. We propose to minimize the following criterion for a good choice of m:

$$g(m) = \Big\|\frac{1}{M}\sum_{s=1}^{M}(\widehat{\mathbf{\Sigma}}_m^{(s)} - \widetilde{\mathbf{\Sigma}}_{2,s})\Big\|_F^2, \tag{4.8}$$

where $\widehat{\mathbf{\Sigma}}_m^{(s)}$ and $\widetilde{\mathbf{\Sigma}}_2^{(s)}$ are defined in (4.5). This criterion is inspired by the one used in Bickel and Levina (2008b) for finding a good banding number, where the true covariance matrix is replaced by a sample one. We demonstrate the performance of the criterion $g(m)$ in section 5 under various settings. Although Theorem 5 suggests m to be such that $m/n \to 1$, finite sample performance may be the best for smaller values of m. Hence, we suggest to search the following split locations in practice in order to minimize $g(m)$:

$$m = [2n^{1/2}, 0.2n, 0.4n, 0.6n, 0.8n, n - 2.5n^{1/2}, n - 1.5n^{1/2}]. \tag{4.9}$$

The smaller splits are actually better when $\mathbf{\Sigma}_p = \sigma^2 \mathbf{I}_p$. This case is excluded in Theorem 5, and we study this by simulations in section 5.

5 Empirical Results

In this section, we demonstrate the performance of our proposed estimator, and compare it with other state-of-the-art estimators under various settings. Hereafter, we abbreviate our method as NERCOME for estimating $\mathbf{\Sigma}_p$ or $\mathbf{\Omega}_p$, as in Nonparametric Eigenvalue-Regularized COvariance Matrix Estimator. The method proposed in Abadir et al. (2014) is abbreviated as CRC (Condition number Regularized Covariance estimator), while the nonlinear shrinkage method in Ledoit and Wolf (2012) is abbreviated as NONLIN. We call the grand average estimator (15) in Abadir et al. (2014) the CRC grand average. Finally, the method of Principal Orthogonal complEment Thresholding in Fan et al. (2013) is abbreviated as POET. We do not include the linear regularization of eigenvalues (also a kind of condition number regularized estimator) in Won et al. (2013), since in Ledoit and Wolf (2012), the performance of NONLIN is already shown to be much better under different degree of dispersion of eigenvalues of $\mathbf{\Sigma}_p$. We also include in some cases the graphical LASSO, abbreviated as GLASSO, as in Friedman et al. (2008), and the adaptive SCAD thresholding, abbreviated as SCAD, which is a special case of POET without any factors.

We create six different profiles of simulations as follows:

(I) Independent data $\mathbf{y}_t \sim N(\mathbf{0}, \mathbf{\Sigma}_p)$, where $\mathbf{\Sigma}_p = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathrm{T}}$. The orthogonal matrix $\mathbf{Q}$ is randomly generated each time, and $\mathbf{D}$ is diagonal with 40% of values being 3, and 60% being 7.

(II) Same as (I), except that $\mathbf{\Sigma}_p = \mathbf{I}_p$.

(III) The data is from a factor model, defined by $\mathbf{y}_t = \mathbf{A}\mathbf{x}_t + \epsilon_t$. The factor loading matrix $\mathbf{A}$ has size $p \times 3$, with elements in $\mathbf{A}$ generated independently from $N(0, 2^2)$. The $\mathbf{x}_t$'s are independently generated from $N(\mathbf{0}, 2\mathbf{I}_r)$, while the $\boldsymbol{\epsilon}_t$'s are independently generated from the standardized t_3 distribution.

(IV) Same as (III), except that the elements in $\mathbf{A}$ are generated independently from $N(0, 3^2)$. The $\mathbf{x}_t$'s are independently generated from $N(\mathbf{0}, 5\mathbf{I}_r)$, while the $\boldsymbol{\epsilon}_t$'s are independently generated from $N(\mathbf{0}, \mathbf{\Sigma}_\epsilon)$.

The matrix $\mathbf{\Sigma}_\epsilon$ has 25% of diagonal values being 2.5, 25% being 2, and 50% being 1.5. The first off-diagonals have all entries being 0.1, and all other off-diagonals are 0.

(V) Same as (IV), except that the $\boldsymbol{\epsilon}_t$'s are independently generated from the standardized t_5 distribution.

(VI) Same as (I), except that $\mathbf{D}$ has 25% of eigenvalues distributed independently as $\tau_i \sim N(3, 0.5^2)$, 25% $\tau_i \sim N(7, 0.5^2)$, 25% being 20 exactly, and 25% $\tau_i \sim N(24, 0.5^2)$.

Profiles (I), (II) and (VI) are cases for non-factor model, with profile (II) being excluded in Theorem 5, since the loss of an ideal estimator is exactly 0 when $\mathbf{\Sigma}_p = \sigma^2\mathbf{I}_p$. Profiles (III) to (V) are for factor models. Profiles (III) and (V) test our method on factor model, since the distribution of $\boldsymbol{\epsilon}_t$ is t_3 and t_5 respectively, which are not allowed in assumption (F1). Profile (IV) has the error covariance matrix changed from diagonal to non-diagonal, with several different values on the main diagonal.

Recall that NONLIN is proved to be asymptotically optimal with respect to the Frobenius and the inverse Stein's loss functions when the data is not from a factor model (see the results in Ledoit and Wolf (2013a) for more details). Hence we expect that NERCOME should be close to NONLIN in performance for profiles (I), (II) and (VI), but better than NONLIN for profiles (III) to (V), since NERCOME is proved to be asymptotically optimal in Theorem 3 for data from a factor model.

We simulate 500 times from the six profiles under all different combinations of $n = 200, 400, 800$ and $p = 50, 100, 200, 500$, and calculate the mean efficiency loss defined in (4.2) with respect to the Frobenius loss and the inverse Stein's loss given in (4.3) and (2.16) respectively. For both NERCOME and CRC, we use $M = 50$ permutations for each split of the data, which gives a good trade-off between accuracy and computational complexity. We use the 7 different split locations as in (4.9) for NERCOME and CRC. We also demonstrate the performance of NERCOME using the automatic choice of m by minimizing $g(m)$ defined in (4.8). The CRC grand average estimator defined in (18) of Abadir et al. (2014) will also be included for comparison, which takes average over the 4 CRC estimators calculated on the split locations $0.2n, 0.4n, 0.6n$ and $0.8n$ respectively.

In the titles of the subsequent graphs, the inverse Stein's loss is abbreviated as the Stein's loss to save space.

Figure 1 shows the mean efficiency loss over 500 simulations for various methods under profile (I). Clearly, the graphical LASSO and the SCAD adaptive thresholding are not performing as good as other methods in terms of the Frobenius loss function, since the covariance and the precision matrices under profile (I) are only approximately sparse. The graphical LASSO has the worst performance, which is to be expected since it concerns with the sparse precision matrix rather than the covariance matrix estimation.

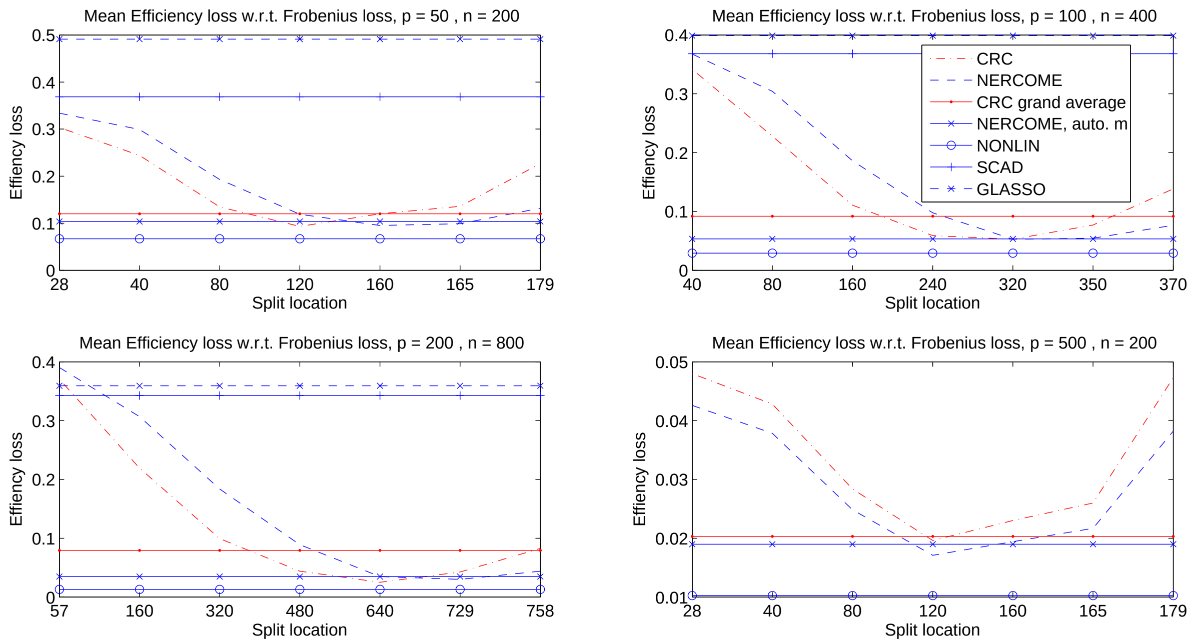

Figure 1: *Mean efficiency loss with respect to the Frobenius loss for profile (I). SCAD and GLASSO are included only when $p < 500$ to save computational time. NERCOME with m automatically chosen is to minimize $g(m)$ in (4.8) over the split locations in (4.9). CRC shares the same split locations. Only NERCOME and CRC are varying over different split locations.*

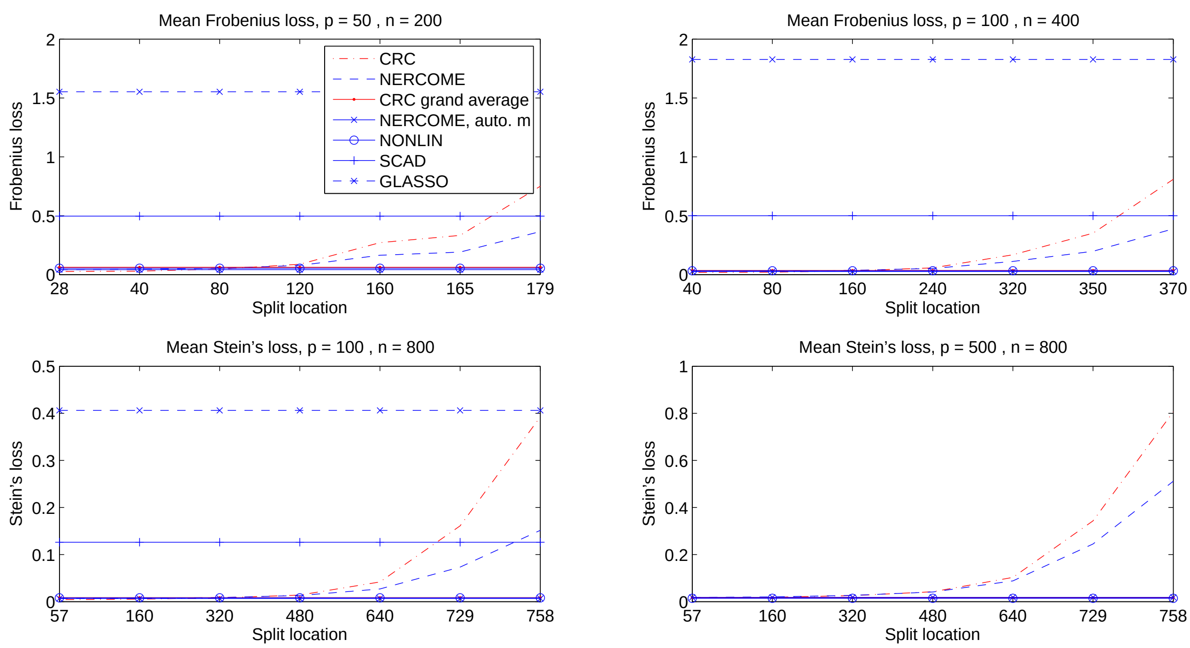

Figure 2: *Mean Frobenius loss (upper row) and inverse Stein's loss (lower row) for profile (II). SCAD and GLASSO are absent for $p < 500$. Refer to the descriptions of Figure 1 for further details.*

NONLIN is the best in all scenarios, followed by NERCOME with m chosen automatically by minimizing (4.8), and CRC grand average. It is fair to say that NONLIN, NERCOME and CRC grand average all perform well in absolute terms. Comparing the performance at different split locations for NERCOME and CRC, we can see that CRC attains a minimum with a smaller split m, and performs better when m is not large. The opposite is true for NERCOME, which usually is at the smallest efficiency loss when m is large. This is consistent with the condition $m/n \to 1$ for asymptotic efficiency in Theorem 5. Clearly, the choice of m that minimizes $g(m)$ in (4.8) is good, since the resulting mean efficiency loss is close to the minimum for NERCOME. We can also see that for CRC, averaging over estimators with split locations from $0.2n$ to $0.8n$, which is the CRC grand average estimator suggested in Abadir et al. (2014), may result in suboptimal performance, since the CRC is usually suboptimal when m is small.

We have omitted the mean efficiency loss with respect to the inverse Stein's loss for profile (I), since the graphs are very similar to the Frobenius loss's counterparts in Figure 1.

Figure 2 shows the actual Frobenius loss and inverse Stein's loss for various methods for profile (II). The efficiency loss in (4.2) is always 1 for any imperfect estimators, since it is easy to see that $\widehat{\mathbf{\Sigma}}_{\text{Ideal}} = \mathbf{\Sigma}_p = \sigma^2\mathbf{I}_p$. Even for $\mathbf{\Sigma}_p = \sigma^2\mathbf{I}_p$, SCAD thresholding and graphical LASSO are not as good as other methods in terms of both loss functions. This can be explained by the fact that even if all the off-diagonal elements of the covariance and the precision matrix estimated from the data are correctly killed off, the diagonal elements are left untouched. Hence, the errors in the diagonal elements will still stack up in the Frobenius or the inverse Stein's loss. On the other hand, CRC, NERCOME and NONLIN are in principal setting the eigenvalues to be as close to $\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i$ as possible. Under $\mathbf{\Sigma}_p = \sigma^2\mathbf{I}_p$, $\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i = \sigma^2$, which is the correct magnitude. The off-diagonal entries will then be close to 0 since $\mathbf{P}\mathrm{diag}(\sigma^2, \ldots, \sigma^2)\mathbf{P}^{\mathrm{T}} = \mathbf{P}(\sigma^2\mathbf{I}_p)\mathbf{P}^{\mathrm{T}} = \sigma^2\mathbf{I}_p$, no matter what orthogonal $\mathbf{P}$ we use. Hence, we can see that the advantage of CRC, NERCOME and NONLIN comes from the fact that the class of estimators $\mathbf{\Sigma}(\mathbf{D}) = \mathbf{P}\mathbf{D}\mathbf{P}^{\mathrm{T}}$ is particularly good when the true covariance matrix is diagonal, with the same entries on the main diagonal throughout. The actual performance of CRC, NERCOME and NONLIN are very close in this case.

It is clear that for both NERCOME and CRC, the best split location is when m is the smallest. This can be seen for NERCOME by noting that, when $\mathbf{\Sigma}_p = \sigma^2\mathbf{I}_p$,

$$\big\|\widehat{\mathbf{\Sigma}}_m - \mathbf{\Sigma}_p\big\|_F^2 = \big\|\mathrm{diag}(\mathbf{P}_1\widetilde{\mathbf{\Sigma}}_2\mathbf{P}_1^{\mathrm{T}}) - \mathbf{P}_1\mathbf{\Sigma}_p\mathbf{P}_1^{\mathrm{T}}\big\|_F^2 = \big\|\mathrm{diag}(\mathbf{P}_1\widetilde{\mathbf{\Sigma}}_2\mathbf{P}_1^{\mathrm{T}}) - \sigma^2\mathbf{I}_p\big\|_F^2 = \sum_{i=1}^{p}(\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i} - \sigma^2)^2.$$

Hence, in order to minimize the loss, we want $\widetilde{\mathbf{\Sigma}}_2$ to be as close to $\sigma^2\mathbf{I}_p$ as possible. It means that $n_2 = n - m$ should be made as large as possible, which implies that m should be as small as possible. The automatic selection for NERCOME clearly performs well even in this case.

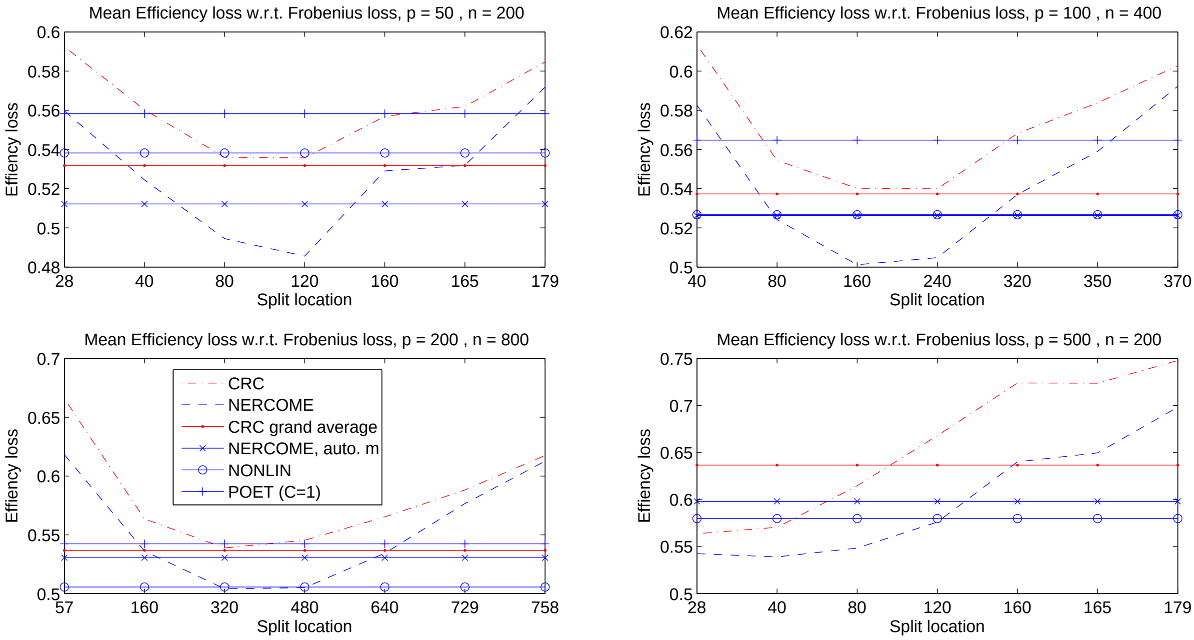

Figure 3: *Mean efficiency loss with respect to the Frobenius loss for profile (III). POET is evaluated at* $C = 0.5$*. NERCOME with* m *automatically chosen is to minimize* $g(m)$ *in (4.8) over the split locations in (4.9). CRC shares the same split locations. Only NERCOME and CRC are varying over different split locations.*

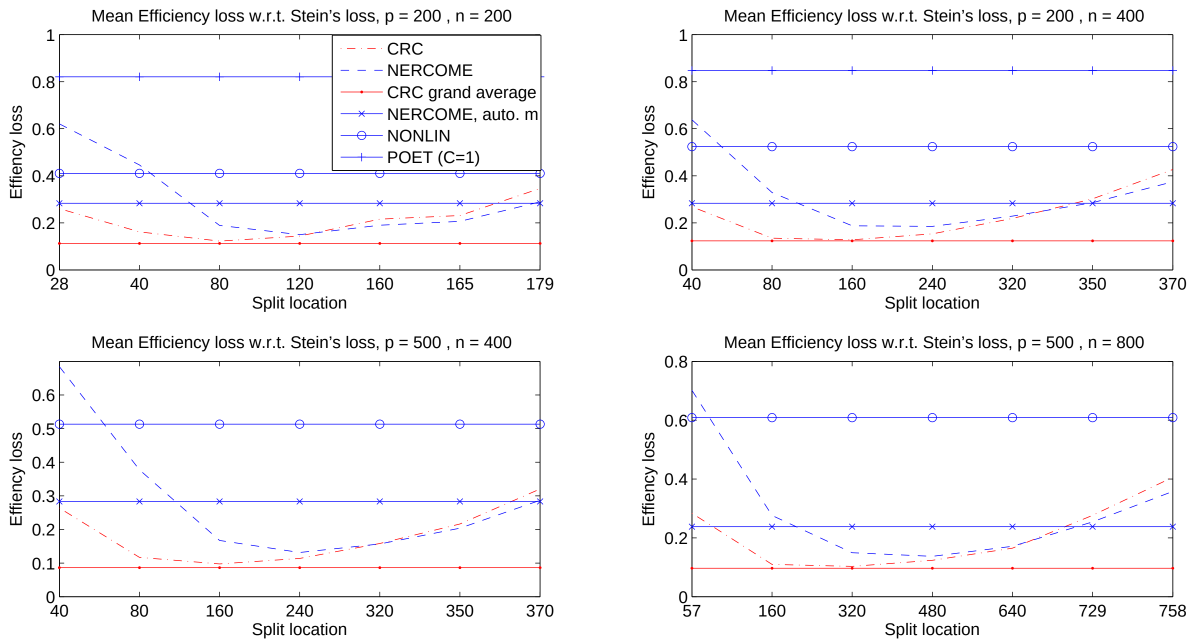

Figure 4: *Mean efficiency loss with respect to the inverse Stein's loss for profile (III). The mean efficiency for NONLIN when* $n = p = 200$ *(upper left panel) is calculated after removing one simulation result with singular covariance matrix. Refer to the descriptions of Figure 3 for further details.*

Figure 3 shows the mean efficiency loss with respect to the Frobenius loss for the factor model in profile (III). All the methods perform badly in absolute terms, as shown by their high efficiency losses. This can be explained by the fact that since the factor model has 3 spiked eigenvalues of order p, any error in estimating those eigenvalues will be hugely reflected in the efficiency loss. The best split location for NERCOME and CRC is in the middle rather than near n as in profile (I). Minimizing $g(m)$ in (4.8) still gives a relatively competitive performance for NERCOME, although it cannot always choose the best split location.

On the other hand, Figure 4 shows a huge improvement for CRC and NERCOME when the inverse Stein's loss is concerned, while POET performs the worst and NONLIN stays at around 50% efficiency loss. In one simulation run for $n = p = 200$, NONLIN returns a singular covariance matrix, and hence the inverse Stein's loss is undefined. This case has to be removed for calculating the mean of efficiency loss. Note that NONLIN always results in a positive semi-definite covariance matrix. However, when $n = p = 200$, it corresponds to $c = 1$ with exactly 1 zero eigenvalue for the sample covariance matrix. This case is not covered in Theorem 4 of Ledoit and Péché (2011), and the fact that the data is from a factor model violates the important assumption in Ledoit and Péché (2011) and Ledoit and Wolf (2013a,b), that we are able to write $\mathbf{y}_t = \boldsymbol{\Sigma}_p^{1/2}\mathbf{z}_t$, with $\mathbf{z}_t$ having independent and identically distributed entries. Hence, although rare in practice, it is not surprising to see a singular covariance matrix estimator for NONLIN. On the other hand, we have proved in Corollary 4 that our estimator is almost surely positive definite even for data from a factor model.

CRC grand average performs the best now, since it appears that all the split locations between $0.2n$ and $0.8n$ are indeed giving good performance to CRC and NERCOME. Part of the reason that POET is not as good as CRC, NERCOME and NONLIN is related to the fact that $\boldsymbol{\Sigma}_\epsilon = \mathbf{I}_p$, which gives advantages to the class of estimators $\boldsymbol{\Sigma}(\mathbf{D}) = \mathbf{PDP}^{\mathrm{T}}$, with arguments similar to the explanations of why SCAD thresholding and graphical LASSO perform worse under profile (II).

To remove such a potential advantage over POET, we create profile (IV), with the error covariance matrix $\boldsymbol{\Sigma}_\epsilon$ now non-diagonal, with several different values on the main diagonal. With respect to the Frobenius loss, there is still a huge efficiency loss for each method and the results are similar to those in Figure 3. Hence we do not show them. Figure 5 shows the results for the inverse Stein's loss. Without the advantage described in the paragraph above for the class of estimators $\boldsymbol{\Sigma}(\mathbf{D}) = \mathbf{PDP}^{\mathrm{T}}$, POET performs extremely well with respect to the inverse Stein's loss. It is even better than the ideal estimator, and hence a negative efficiency loss (see the bottom right panel). The other 3 panels in Figure 5 have omitted POET so that the performance of other methods are clearer for comparison. While CRC grand average and NERCOME perform closely and well in general (apart from POET), NONLIN are performing much worse than other methods. This is expected, since NONLIN is not applicable for factor models. For the case when $n = p = 200$, NONLIN suffers from producing singular covariance matrix estimators again, as in profile (III).

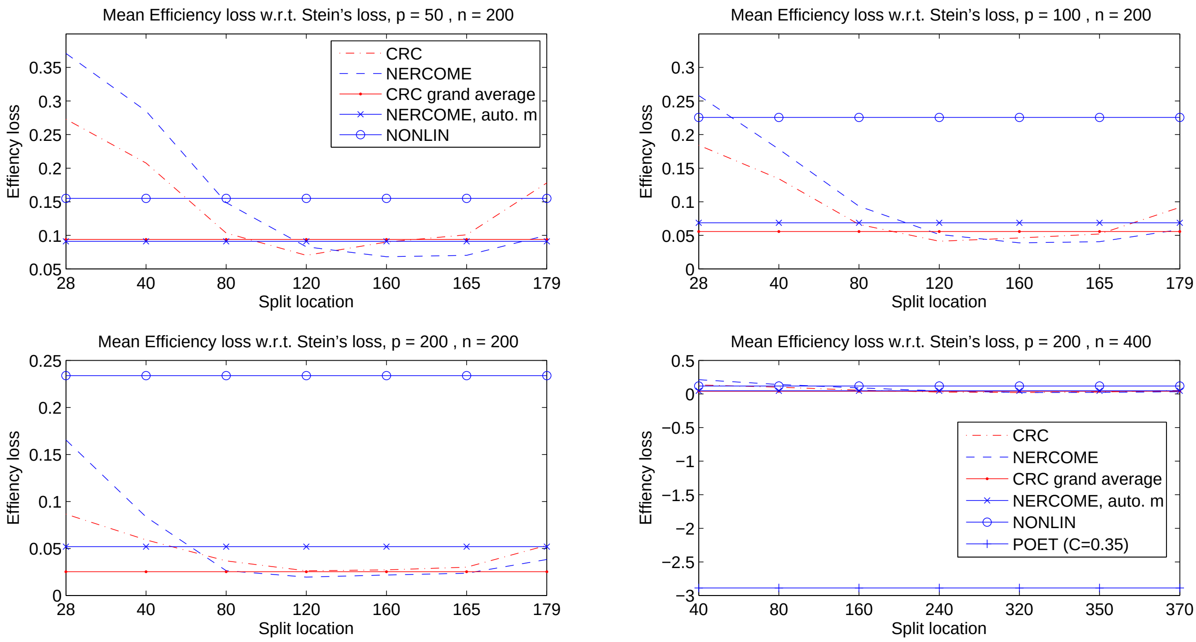

Figure 5: *Mean efficiency loss with respect to the inverse Stein's loss for profile (IV). The mean efficiency for NONLIN when $n = p = 200$ (lower left panel) is calculated after removing 4 simulation results with singular covariance matrix estimators. Refer to the descriptions of Figure 3 for further details*

Figure 6 shows the mean efficiency loss for profile (V). We see that NERCOME and CRC grand average are performing well despite the error distribution being t_5 and violating the moment assumptions in (F1), with CRC grand average being slightly better in performance than NERCOME with automatic choice of split. POET and NONLIN are not performing well on the other hand, with reasons similar to those in the explanations of Figure 4.

Figure 7 shows the results for profile (VI) simulations. We can see that NONLIN is the best, followed closely by NERCOME with automatically chosen m, showing that minimizing $g(m)$ in (4.8) does give a good choice of m for NERCOME. With respect to the Frobenius loss or the inverse Stein's loss, we see that both CRC and NERCOME always achieve suboptimal performance over the split locations $0.2n$ and $0.4n$ under a wide variety of (n, p) combinations. This explains why the CRC grand average is not performing as good as NERCOME and NONLIN, although the minimum mean efficiency loss for CRC is comparable to NERCOME in cases where $p < n$. When $p > n$, the performance of NERCOME dominates CRC over all split locations, as shown in the lower left panel of Figure 7.

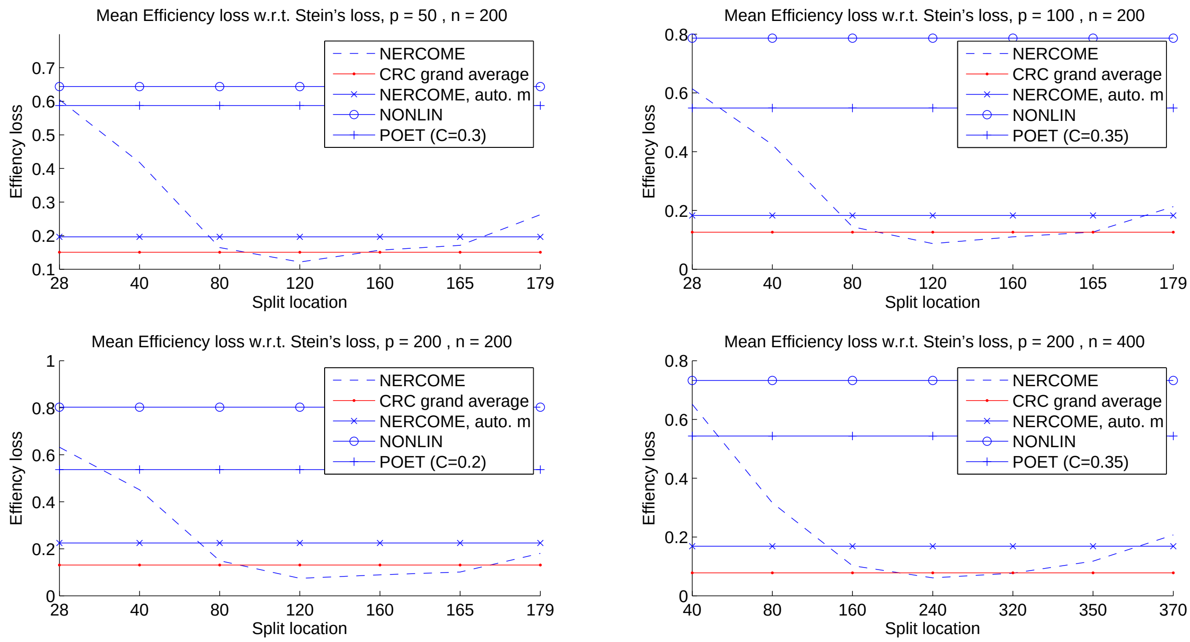

Figure 6: *Mean efficiency loss with respect to the inverse Stein's loss for profile (V). Refer to the descriptions of Figure 3 for further details*

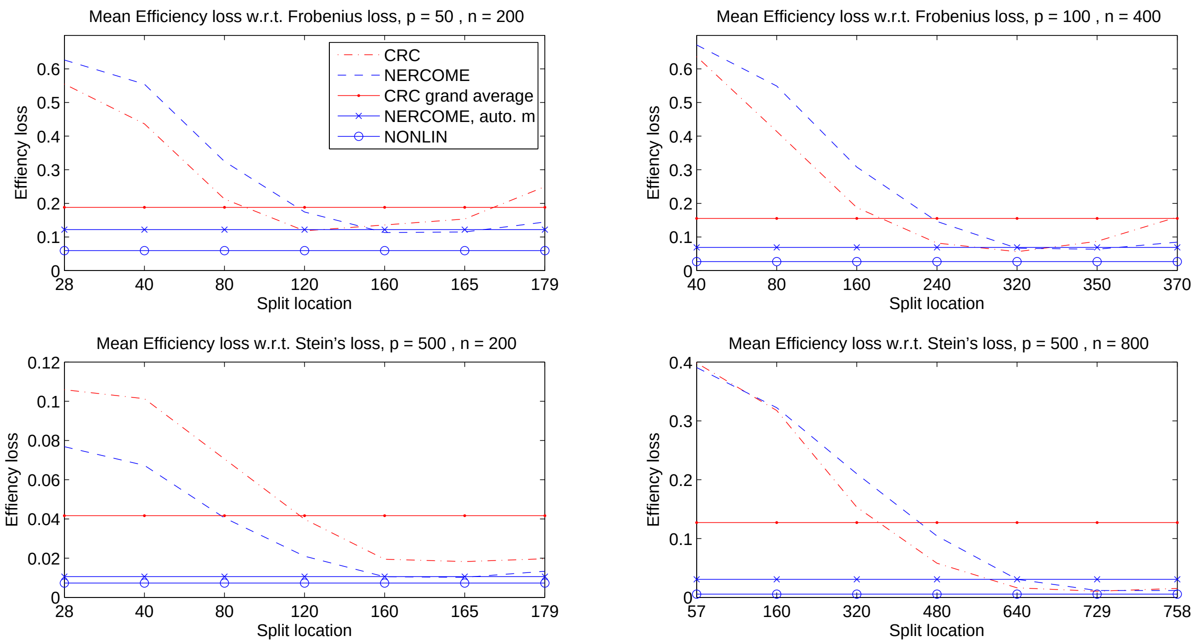

Figure 7: *Mean Frobenius loss (upper row) and inverse Stein's loss (lower row) for profile (VI).*

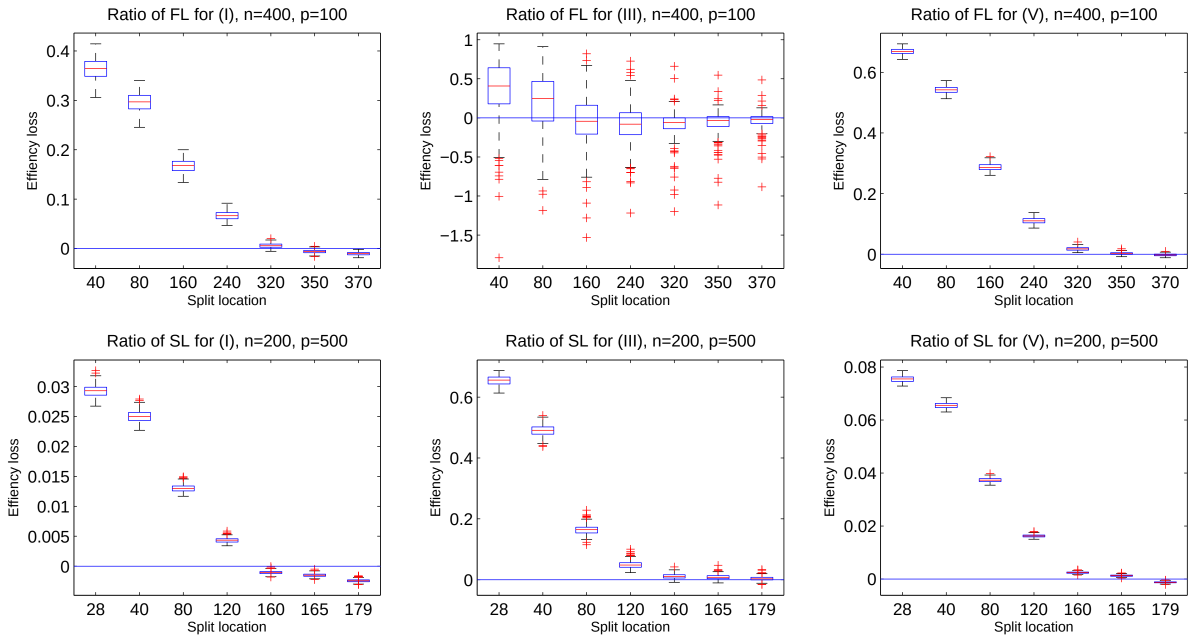

Figure 8: *Boxplots of efficiency loss of* $M^{-1}\sum_{i=1}^{M}\widehat{\mathbf{\Sigma}}^{(i)}_{\text{Ideal},m}$ *at split locations defined in (4.9). Upper row: With respect to the Frobenius loss, with* $n = 400, p = 100$*. Lower row: With respect to the inverse Stein's loss, with* $n = 200, p = 500$*. Left panel : For profile (I). Middle panel : For profile (III). Right panel: For profile (VI).*

Figure 8 shows the boxplots of the efficiency loss of the averaged ideal estimator $\widehat{\mathbf{\Sigma}}_{\text{Ideal},m,M} = M^{-1}\sum_{i=1}^{M}\widehat{\mathbf{\Sigma}}^{(i)}_{\text{Ideal},m}$ for various profiles. We show this because it is important for the ratio of loss $L(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_{\text{Ideal},m,M})/L(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_{\text{Ideal}})$ to be going to 1 almost surely (i.e., $EL(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_{\text{Ideal},m,M}) \xrightarrow{a.s.} 0$, which is what Figure 8 is showing) to ensure our estimator $\widehat{\mathbf{\Sigma}}_{m,M}$ to be almost surely asymptotically efficient. See Lemma 5 in the Appendix also for more details. From the figure, it is clear that for the profiles (I), (III) and (VI), the split location m should be close to n for $EL(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_{\text{Ideal},m,M})$ to be close to 0. Since results from profile (IV) and (V) are similar to (III), we choose not to show all the results here. The efficiency loss has a wide variability for profile (III) with respect to the Frobenius loss (the upper middle panel of Figure 8). It certainly achieves 0 on average for several split locations not close to n, so that the best m is not always close to n for $\widehat{\mathbf{\Sigma}}_{m,M}$, as shown in Figure 3.

For profile (I), with negative efficiency loss in all 500 simulation runs for the averaged ideal estimator $\widehat{\mathbf{\Sigma}}_{\text{Ideal},m,M}$ when split location is close to n (see the left panels of Figure 8), it is clear that using the eigenmatrix $\mathbf{P}_1$ to form our estimator and averaging over different permutations of the data can be better than just using $\mathbf{P}$ from the full data set. Our estimator $\widehat{\mathbf{\Sigma}}_{m,M}$ in (4.6) follows the same spirit exactly.

Finally, we compare the computing time for NERCOME, NONLIN, SCAD and GLASSO for profile (I).

		$p = 50$	$p = 100$	$p = 200$	$p = 500$
$n = 200$	NERCOME	$.43_{(.0)}$	$1.8_{(.2)}$	$8.4_{(.4)}$	$78.3_{(2.6)}$
	NONLIN	$21.4_{(1.3)}$	$29.3_{(6.0)}$	$39.0_{(5.9)}$	$38.9_{(7.4)}$
	SCAD	$.1_{(.0)}$	$.7_{(.0)}$	$4.8_{(.0)}$	-
	GLASSO	$6.5_{(.9)}$	$25.1_{(4.2)}$	$93.2_{(3.3)}$	-
n=400	NERCOME	$.50_{(.0)}$	$2.3_{(.1)}$	$9.9_{(.5)}$	$70.5_{(9.5)}$
	NONLIN	$25.3_{(12.0)}$	$33.4_{(10.1)}$	$38.4_{(3.5)}$	$55.1_{(10.1)}$
	SCAD	$.2_{(.0)}$	$1.2_{(.0)}$	$8.6_{(.1)}$	-
	GLASSO	$6.3_{(.2)}$	$31.9_{(4.4)}$	$104.6_{(4.6)}$	-
n=800	NERCOME	$.64_{(.0)}$	$2.7_{(.1)}$	$11.7_{(.6)}$	$76.4_{(25.4)}$
	NONLIN	$24.8_{(9.8}$	$34.0_{(12.1)}$	$40.0_{(8.4)}$	$83.6_{(135.9)}$
	SCAD	$.3_{(.0)}$	$2.2_{(.1)}$	$16.3_{(.4)}$	-
	GLASSO	$6.3_{(.3)}$	$31.7_{(.8)}$	$150.8_{(454.8)}$	-

Table 1: *Mean time (in seconds) for computing a covariance matrix estimator for* NERCOME *(including the time for finding the best split using (4.8)),* NONLIN*, SCAD and GLASSO for profile (I). Standard deviation is in bracket.*

While all methods are Matlab coded, only NONLIN requires a third-party SLP optimizer, since NONLIN involves solving nonconvex optimization problems, which is done using a commercial package called SNOPT in Matlab (see Ledoit and Wolf (2012) for more details). From table 1, it is clear that SCAD thresholding is the fastest, albeit we have set favorable values of C for the thresholding in advance. NERCOME is the second fastest for $p \leq 200$. When $p \geq 500$, NONLIN is faster than NERCOME for smaller values of n, but the computational cost increases quickly with the increase of n. NERCOME, on the other hand, remains similar in computational costs for a wide range of values of n. This is because the major computational cost for NERCOME comes from the M eigen-decompositions of a $p \times p$ sample covariance matrix, with each eigen-decomposition being computationally expensive when p is large. Increasing n only marginally increases the computational cost of an eigen-decomposition.

5.1 Application: Bias reduction with generalized least squares (GLS)

Consider a linear model

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \boldsymbol{\epsilon}_i, \quad i = 1, \dots, n,$$

where the $\mathbf{X}_i$'s are known covariates and the $\boldsymbol{\epsilon}_i$'s have $E(\boldsymbol{\epsilon}_i\boldsymbol{\epsilon}_i^{\mathrm{T}}) = \boldsymbol{\Sigma}_p$. Then the generalized least squares estimator

$$\widehat{\boldsymbol{\beta}}_{\mathrm{GLS}} = \left(\sum_{i=1}^{n} \mathbf{X}_i^{\mathrm{T}}\boldsymbol{\Sigma}_p^{-1}\mathbf{X}_i\right)^{-1} \sum_{i=1}^{n} \mathbf{X}_i^{\mathrm{T}}\boldsymbol{\Sigma}_p^{-1}\mathbf{y}_i$$

is more efficient than the least squares one in general . However, we need the precision matrix $\boldsymbol{\Sigma}_p^{-1}$ as an input. We demonstrate the effectiveness of bias reduction of $\widehat{\boldsymbol{\beta}}_{\mathrm{GLS}}$ using different methods in estimating $\boldsymbol{\Sigma}_p^{-1}$.

To this end, we run 500 simulations. We set $\boldsymbol{\beta} = (-0.5, 0.5, 0.3, -0.6)^{\mathrm{T}}$. In each simulation run, we generate $\mathbf{X}_i$ with independent $N(0, 1)$ entries. For the noise, we use the standardized macroeconomic data

$\mathbf{w}_i$ analyzed in Stock and Watson (2005). The data consists of $p = 132$ monthly U.S. macroeconomic time series running from January 1959 to December 2003 ($n = 526$), and is categorized into 14 categories of different size. In Stock and Watson (2005), they argue that there are 7 factors in the data. We set $\boldsymbol{\epsilon}_i = 2\mathbf{w}_i + \mathbf{z}_i$, where $\mathbf{z}_i$ consists of independent $N(0, 0.2^2)$ entries and they are generated in each simulation run. Hence, we have $\mathbf{y}_t = \mathbf{X}_t\boldsymbol{\beta} + 2\mathbf{w}_i + \mathbf{z}_i$. We choose to use $\boldsymbol{\epsilon}_i = 2\mathbf{w}_i + \mathbf{z}_i$ in order to add challenges to NERCOME, as it does not only contain potentially many factors, but also exhibits certain degree of serial correlation, which is violating our assumptions of independence. For NERCOME, we choose m to minimize (4.8) on a grid of 7 split locations as in (4.9).

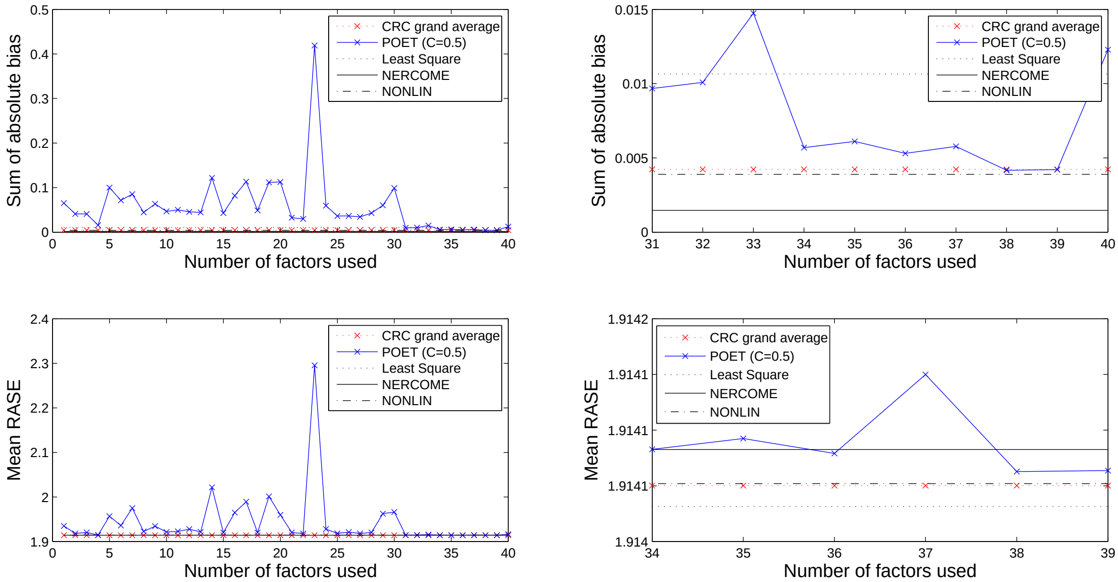

Figure 9: *Upper row: Mean sum of absolute bias for estimating $\boldsymbol{\beta}$ against the number of factors K for POET. Lower row: The mean root-average-square prediction error against K. Least squares, NERCOME, CRC grand average and NONLIN stay constant throughout. Left: Number of factors used in POET from 1 to 40. Right: Number of factors used in POET from 31 to 40 (upper right) or from 34 to 39 (lower right).*

Figure 9 shows the mean sum of absolute bias $\|\widehat{\boldsymbol{\beta}}_{\mathrm{GLS}} - \boldsymbol{\beta}\|_1$, as well as the mean root-average-square prediction error (RASE) $n^{-1}\sum_{t=1}^{n} p^{-1/2}\|\mathbf{y}_t - \mathbf{X}_t\widehat{\boldsymbol{\beta}}_{\mathrm{GLS}}\|$, for different methods. In Fan et al. (2013), POET is described as being sensitive to underestimating the number of factors. This is indeed the case, as both the upper and lower left panels in the figure show a lot of fluctuations as the number of factors K used in POET varies, until around $K \geq 31$. This indicates that the number of factors in $\mathbf{w}_i$ is likely to be around 31, rather than 7 as suggested in Stock and Watson (2005).

The upper right panel of the figure shows the sum of absolute bias when $K \geq 31$. Clearly, all the methods can improve upon the least squares one, which incurs the largest bias. CRC grand average, NONLIN and POET at $K = 38, 39$ have similar performance, while NERCOME has the best performance throughout. Nevertheless, apart from the least squares, all of the other methods perform well in absolute terms.

The lower right panel of the figure shows the mean root-average-square prediction error when $K = 34, \ldots, 39$. This time, the least squares method incurs the smallest error, followed by CRC grand average and NONLIN. NERCOME and POET perform slightly worse than other methods. Again, all the methods perform well in absolute terms, especially the difference in errors are actually very small.

5.2 Application: Risk minimization of a portfolio

We consider risk minimization for a portfolio of $p = 100$ stocks, which is also considered in section 7.2 of Fan et al. (2013). The data consists of 2640 annualized daily excess returns $\{r_t\}$ for the period January 1st 2001 to December 31st 2010 (22 trading days each month). Five portfolios are created at the beginning of each month using five different methods in estimating the covariance matrix of returns. A typical setting here is $n = 264, p = 100$, that is, one year of past returns to estimate a covariance matrix of 100 stocks. Each portfolio has weights given by

$$\widehat{\mathbf{w}} = \frac{\widehat{\boldsymbol{\Sigma}}^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}^{-1}\mathbf{1}_p},$$

where $\mathbf{1}_p$ is the vector of p ones, and $\widehat{\boldsymbol{\Sigma}}^{-1}$ is an estimator of the $p \times p$ precision matrix for the stock returns, using strict factor model (covariance of error forced to be diagonal, abbreviated as SFM), POET (constant C determined by cross-validation), CRC grand average, NERCOME (m automatically chosen by minimizing (4.8)) and NONLIN respectively. This weight formula solves the risk minimization problem

$$\min_{\mathbf{w}:\mathbf{w}^{\mathrm{T}}\mathbf{1}_p=1} \mathbf{w}^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}\mathbf{w}.$$

At the end of each month, for each portfolio, we compute the portfolio return, the out-of-sample variance and the Sharpe ratio, given respectively by (see also Demiguel and Nogales (2009))

$$\widehat{\mu}_i = \frac{1}{22}\sum_{t=22i+1}^{22i+22}\mathbf{w}^{\mathrm{T}}\mathbf{r}_t, \quad \widehat{\sigma}_i^2 = \frac{1}{21}\sum_{t=22i+1}^{22i+22}(\mathbf{w}^{\mathrm{T}}\mathbf{r}_t - \widehat{\mu}_i)^2, \quad \widehat{\mathrm{sr}} = \frac{\widehat{\mu}_i}{\widehat{\sigma}_i^2}, \quad i = 12, \ldots, 119.$$

Table 2 shows the results. In terms of risk minimization, the strict factor model is not as good as NERCOME, as NERCOME has a relatively large overall risk improvement. In terms of proportion of better performance, POET is the best, but NERCOME has a slight overall risk improvement over POET, and has also a larger risk improvement when outperforms. CRC grand average does not perform as well

Out-of-sample variance	SFM	POET	CRC	NONLIN
Proportion outperforms	57.4%	47.2%	58.3%	49.1%
Mean risk improvement	5.0%	0.4%	0.0%	-0.0%
Mean risk improvement when outperforms	12.1%	6.8%	1.3%	1.7%
Mean extra risk when performing worse	4.5%	5.4%	1.9%	1.7%
Portfolio return				
Proportion outperforms	48.2%	58.3%	47.2%	50%
Mean return improvement	-1.15%	0.7%	-0.0%	0.0%
Mean return improvement when outperforms	4.7%	3.3%	0.7%	1.1%
Mean return shortfall when performing worse	6.6%	2.9%	0.7%	1.0%
Sharpe ratio				
Proportion outperforms	50.0%	55.6%	50%	49.1%
Mean improvement	-1.4%	1.3%	0.2%	0.5%
Mean improvement when outperforms	11.9%	8.2%	2.1%	3.0%
Mean shortfall when performing worse	14.8%	7.3%	1.7%	2.0%

Table 2: *Performance of different methods relative to* NERCOME. SFM represents the strict factor model, with diagonal covariance matrix. *First row: Proportion of months (108 of them) when* NERCOME *has a smaller risk, $\widehat{\sigma}_i^2$. Second row: Mean of risk improvement in percentage points. Third row: Same, but only when* NERCOME *outperforms. Fourth row: same, but only when* NERCOME *performs worse. Similar interpretations for Portfolio return and Sharpe ratio.*

as NERCOME, while NONLIN is slightly better. It is fair to say that POET, NERCOME and NONLIN are all performing well for slightly different reasons, while NERCOME and NONLIN are slightly better overall.

In terms of portfolio return and sharpe ratio, the strict factor model is the best overall, while POET is the worst. NERCOME, CRC grand average and NONLIN all perform similarly.

6 Conclusion

We investigate the problem of high dimensional covariance matrix estimation through regularizing the eigenvalues. Considering the class of rotation-equivariant covariance matrix, we split the data into two parts, and obtain a regularization of eigenvalues which is proved to be asymptotically optimal in the sense of minimizing the Frobenius loss. Incidentally, the inverse of this estimator is also asymptotically optimal in estimating the precision matrix with respect to the inverse Stein's loss. We also proved that our estimator is almost surely positive definite, and is asymptotically efficient relative to an ideal estimator which assumes the knowledge of the true covariance matrix itself.

The method proposed is applicable to data from a factor model also, which considerably expands its scope of applicability. When estimating the covariance or the precision matrix for data from a factor model, one usually needs to estimate the number of unknown factors, the factors themselves and the factor loading matrix. Our method bypasses this step, and gives an estimator of the covariance and the precision matrix directly. We find that our estimator still enjoy the aforementioned asymptotic optimality. We can see from the simulation results that the performance of the proposed estimator is very good and is comparable or better than other state-of-the-art methods.

7 Appendix

We state an important Lemma first, which is in fact Lemma 2.7 of Bai and Silverstein (1998). Then, we present two more lemmas before presenting the proof of Theorem 1.

Lemma 2. *Let $\mathbf{y} = (y_1, \ldots, y_p)^{\mathrm{T}}$ be a complex random vector with independent and identically distributed (i.i.d.) entries satisfying $E(y_1) = 0$, $E|y_1|^2 = 1$. Let $\mathbf{A}$ be a given $p \times p$ matrix with possibly complex entries. Then for any $q \geq 2$,*

$$E|\mathbf{y}^{\mathrm{T}}\mathbf{A}\mathbf{y} - tr(\mathbf{A})|^q \leq K_q \left(E^{q/2}|y_1|^4 \mathrm{tr}^{q/2}(\mathbf{A}\mathbf{A}^*) + E|y_1|^{2q}\mathrm{tr}(\mathbf{A}\mathbf{A}^*)^{q/2} \right),$$

where K_q is a constant depending only on q, and $\mathbf{A}^* = \bar{\mathbf{A}}^{\mathrm{T}}$ is the Hermitian of $\mathbf{A}$.

Lemma 3. *For $z \in \mathbb{C}^+$, we have $\|R_1(z)\|^2 \leq \frac{1}{\mathrm{Im}^2(z)}$.*

Proof of Lemma 3. Denote $\overline{z}$ the conjugate of z, and let $\widetilde{\boldsymbol{\Sigma}}_1 = \mathbf{P}_1\mathbf{D}_1\mathbf{P}_1^{\mathrm{T}}$. Then

$$\begin{aligned}
\|R_1(z)\|^2 &= \lambda_{\max}(R_1(z)\overline{R_1(z)}) \\
&= \lambda_{\max}(\mathbf{P}_1(\mathbf{D}_1 - z\mathbf{I}_p)^{-1}\mathbf{P}_1^{\mathrm{T}}\mathbf{P}_1(\mathbf{D}_1 - \overline{z}\mathbf{I}_p)^{-1}\mathbf{P}_1^{\mathrm{T}}) \\
&= \lambda_{\max}\left(\{[\mathbf{D}_1 - \mathrm{Re}(z)]^2 + \mathrm{Im}^2(z)\}^{-1}\right) \\
&\leq \frac{1}{\mathrm{Im}^2(z)}. \quad \square
\end{aligned}$$

Lemma 4. *Let $\mathbf{R}$ be Hermitian, that is, it is symmetric with complex entries. Then for a real square matrix $\mathbf{B}$, we have $|\mathrm{tr}(\mathbf{B}\mathbf{R})| \leq |\lambda_{\max}(\mathbf{B})||\mathrm{tr}(\mathbf{R})|$. Also, $|\mathrm{tr}(\mathbf{A}^{\mathrm{T}}\mathbf{R}\mathbf{A})| \leq 2^{1/2}\|\mathbf{R}\|\mathrm{tr}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ for a real matrix $\mathbf{A}$.*

Proof of Lemma 4. Since $\mathbf{R}$ is Hermitian, it can be decomposed as $\mathbf{R} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathrm{T}}$, where $\mathbf{U}$ is orthogonal and $\mathbf{D}$ is diagonal with complex entries. Writing $\mathbf{U} = (\mathbf{u}_1, \ldots, \mathbf{u}_p)$ and $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_p)$, then

$$|\mathrm{tr}(\mathbf{B}\mathbf{R})| = |\mathrm{tr}(\mathbf{U}^{\mathrm{T}}\mathbf{B}\mathbf{U}\mathbf{D})| = \left|\sum_{i=1}^{p} d_i \mathbf{u}_i^{\mathrm{T}}\mathbf{B}\mathbf{u}_i\right| \leq |\lambda_{\max}(\mathbf{B})| \cdot \left|\sum_{i=1}^{p} d_i\right| = |\lambda_{\max}(\mathbf{B})||\mathrm{tr}(\mathbf{R})|.$$

Finally, writing $\mathbf{A} = (\mathbf{a}_1, \dots, \mathbf{a}_r)$,

$$
\begin{aligned}
|\mathrm{tr}(\mathbf{A}^{\mathrm{T}}\mathbf{R}\mathbf{A})| = \left|\sum_{i=1}^{r} \mathbf{a}_i^{\mathrm{T}}\mathbf{R}\mathbf{a}_i\right| &= \left(\left[\sum_{i=1}^{r} \mathbf{a}_i^{\mathrm{T}}\mathrm{Re}(\mathbf{R})\mathbf{a}_i\right]^2 + \left[\sum_{i=1}^{r} \mathbf{a}_i^{\mathrm{T}}\mathrm{Im}(\mathbf{R})\mathbf{a}_i\right]^2\right)^{1/2} \\
&\le \left(\lambda_{\max}^2[\mathrm{Re}(\mathbf{R})] + \lambda_{\max}^2[\mathrm{Im}(\mathbf{R})]\right)^{1/2} \mathrm{tr}(\mathbf{A}^{\mathrm{T}}\mathbf{A}) \\
&\le \left(2\lambda_{\max}[\mathrm{Re}^2(\mathbf{R}) + \mathrm{Im}^2(\mathbf{R})]\right)^{1/2} \mathrm{tr}(\mathbf{A}^{\mathrm{T}}\mathbf{A}) = 2^{1/2}\big\|\mathbf{R}\big\|\mathrm{tr}(\mathbf{A}^{\mathrm{T}}\mathbf{A}). \quad \square
\end{aligned}
$$

Proof of Theorem 1. We prove part (i) first. Consider $\mathbf{\Psi}_m^{(1)}(z) - \mathbf{\Psi}^{(1)}(z) = I_1 + I_2$, where

$$
I_1 = \mathbf{\Psi}_m^{(1)}(z) - \frac{1}{p}\mathrm{tr}\left(R_1(z)\mathbf{\Sigma}_p\right), \quad I_2 = \frac{1}{p}\mathrm{tr}\left(R_1(z)\mathbf{\Sigma}_p\right) - \mathbf{\Psi}^{(1)}(z),
$$

with $R_1(z) = (\widetilde{\mathbf{\Sigma}}_1 - z\mathbf{I}_p)^{-1}$. A direct application of Lemma 2 of Ledoit and Péché (2011) implies that $p^{-1}\mathrm{tr}(R_1(z)\mathbf{\Sigma}_p)$ converges a.s. to $\mathbf{\Psi}^{(1)}(z)$, and hence $I_2 \overset{a.s.}{\to} 0$. It remains to show that $I_1 \overset{a.s.}{\to} 0$ as well.

To this end, write $\mathbf{Y}_2 = (\mathbf{y}_{21}, \dots, \mathbf{y}_{2n_2})$, so that $\widetilde{\mathbf{\Sigma}}_2 = n_2^{-1}\sum_{k=1}^{n_2} \mathbf{y}_{2k}\mathbf{y}_{2k}^{\mathrm{T}}$. Also, write $\mathbf{y}_{2k} = \mathbf{\Sigma}_p^{1/2}\mathbf{z}_{2k}$ for $k = 1, \dots, n_2$ with $\mathbf{z}_{2k}$ being one of the $\mathbf{z}_i$'s in assumption (A1). Using the independence of the $\mathbf{z}_{2k}$'s and $\mathbf{Y}_1$, we have for $k = 1, \dots, n_2$, $2 \le q \le 6$ and $z \in \mathbb{C}^+$,

$$
\begin{aligned}
&E\left\{\left|\mathbf{z}_{2k}^{\mathrm{T}}\mathbf{\Sigma}_p^{1/2}R_1(z)\mathbf{\Sigma}_p^{1/2}\mathbf{z}_{2k} - \mathrm{tr}(\mathbf{\Sigma}_p^{1/2}R_1(z)\mathbf{\Sigma}_p^{1/2})\right|^q \Big|\mathbf{Y}_1\right\} \\
&\le K_q(E^{q/2}|z_{2k}|^4\mathrm{tr}^{q/2}(\mathbf{\Sigma}_pR_1(z)\mathbf{\Sigma}_p\overline{R_1(z)}) + E|z_{2k}|^{2q}\mathrm{tr}(\mathbf{\Sigma}_p^{1/2}R_1(z)\mathbf{\Sigma}_p\overline{R_1(z)}\mathbf{\Sigma}_p^{1/2})^{q/2}) \\
&\le K_q(E^{q/2}|z_{2k}|^4\big\|\mathbf{\Sigma}_p\big\|^q p^{q/2}\mathrm{Im}^{-q}(z) + E|z_{2k}|^{2q}p\lambda_{\max}^{q/2}(\mathbf{\Sigma}_p^{1/2}R_1(z)\mathbf{\Sigma}_p\overline{R_1(z)}\mathbf{\Sigma}_p^{1/2})) \\
&\le K_p\mathrm{Im}^{-q}(z)p^{q/2}\big\|\mathbf{\Sigma}_p\big\|^q(E^{q/2}|z_{2k}|^4 + p^{1-q/2}E|z_{2k}|^{2q}) = O(p^q),
\end{aligned}
$$

where the second line uses Lemma 2, which is applicable by assumption (A1) on $\mathbf{z}_{2k}$, and the second last line is by Lemma 4. The last line uses the assumptions in (A1). We have

$$
\begin{aligned}
E(|I_1|^6|\mathbf{Y}_1) &= E\left(\left|\frac{1}{pn_2}\mathrm{tr}\left(R_1(z)\sum_{k=1}^{n_2}\mathbf{y}_{2k}\mathbf{y}_{2k}^{\mathrm{T}}\right) - \frac{1}{p}\mathrm{tr}(R_1(z)\mathbf{\Sigma}_p)\right|^6 \Big|\mathbf{Y}_1\right) \\
&= E\left(\left|\frac{1}{pn_2}\sum_{k=1}^{n_2}(\mathbf{z}_{2k}^{\mathrm{T}}\mathbf{\Sigma}_p^{1/2}R_1(z)\mathbf{\Sigma}_p^{1/2}\mathbf{z}_{2k} - \mathrm{tr}(R_1(z)\mathbf{\Sigma}_p))\right|^6 \Big|\mathbf{Y}_1\right).
\end{aligned}
$$

Define $g_k = \mathbf{z}_{2k}^{\mathrm{T}}\mathbf{\Sigma}_p^{1/2}R_1(z)\mathbf{\Sigma}_p^{1/2}\mathbf{z}_{2k} - \mathrm{tr}(R_1(z)\mathbf{\Sigma}_p)$. Then $E(g_i|\mathbf{Y}_1) = 0$, and $g_i|\mathbf{Y}_1$ is independent of $g_j|\mathbf{Y}_1$

for $i \neq j$. We then have the expansion

$$\begin{aligned}
E(|I_1|^6) &= \frac{1}{p^6 n_2^6} \sum_{i_1,\ldots,i_6=1}^{n_2} E(E(g_{i_1}\bar{g}_{i_2} g_{i_3}\bar{g}_{i_4} g_{i_5}\bar{g}_{i_6}|\mathbf{Y}_1)) \\
&\leq \frac{1}{p^6 n_2^6} E\bigg(\sum_{i=1}^{n_2} E(|g_i|^6|\mathbf{Y}_1) + \sum_{i\neq j} [E(|g_i|^2|\mathbf{Y}_1)E(|g_j|^4|\mathbf{Y}_1) + E(|g_i|^3|\mathbf{Y}_1)E(|g_j|^3|\mathbf{Y}_1)] \\
&\qquad + \sum_{i\neq j\neq k} E(|g_i|^2|\mathbf{Y}_1)E(|g_j|^2|\mathbf{Y}_1)E(|g_k|^2|\mathbf{Y}_1)\bigg) \qquad (7.1)\\
&= O(p^{-6}n_2^{-6}\cdot[n_2 + 2n_2(n_2-1) + n_2(n_2-1)(n_2-2)]p^6) = O(n_2^{-3}),
\end{aligned}$$

where the last line used $E(|g_i|^q|\mathbf{Y}_1) = O(p^q)$ proved before. Since $\sum_{n\geq 1} n_2^{-3} < \infty$ by assumption, we can apply the Borel-Cantelli lemma to conclude that $I_1 \stackrel{a.s.}{\to} 0$. This completes the proof of part (i).

Note also that since $I_1 \stackrel{a.s.}{\to} 0$, we have $\mathbf{\Psi}_m^{(1)}(z) - p^{-1}\mathrm{tr}(R_1(z)\mathbf{\Sigma}_p) \stackrel{a.s.}{\to} 0$. Using the inversion formula (2.3), it is easy to show that the inverse Stieltjes transform for $\mathbf{\Psi}_m^{(1)}(z)$ is $\mathbf{\Phi}_m^{(1)}(x)$ in (2.13) on all points of continuity of $\mathbf{\Phi}_m^{(1)}(x)$, and that for $p^{-1}\mathrm{tr}(R_1(z)\mathbf{\Sigma}_p)$ is $p^{-1}\sum_{i=1}^p \mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}\mathbf{1}_{\{\lambda_{1i}\leq x\}}$ on all points of continuity of the function. Hence, using equation (2.5) of Silverstein and Bai (1995), we conclude that

$$\mathbf{\Phi}_m^{(1)}(x) - \frac{1}{p}\sum_{i=1}^p \mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}\mathbf{1}_{\{\lambda_{1i}\leq x\}} \stackrel{a.s.}{\to} 0$$

as $n_1, p \to \infty$ with $p/n_1 \to c_1 > 0$, which certainly include the case $c_1 = 1$. This proves the last part of the theorem.

The proof for part (ii) of the theorem can be found in the proof of Theorem 4 of Ledoit and Péché (2011), from equations (45) to (50) by replacing γ there with $1/c_1$ and F with F_1. Part (iii) is just an application of Theorem 4 of Ledoit and Péché (2011). This completes the proof of the theorem. □

Proof of Theorem 3. Let $\mathbf{Y}_2 = (\mathbf{y}_{21}, \ldots, \mathbf{y}_{2n_2})$ with $\mathbf{y}_{2k} = \mathbf{A}\mathbf{x}_{2k} + \boldsymbol{\epsilon}_{2k}$, $k = 1, \ldots, n_2$. Let $R_1(z) = (\widetilde{\mathbf{\Sigma}}_1 - z\mathbf{I}_p)^{-1}$. Then with $\mathbf{\Sigma}_p = \mathbf{A}\mathbf{\Sigma}_x\mathbf{A} + \mathbf{\Sigma}_\epsilon$ as in (3.18), we have

$$\begin{aligned}
&\frac{1}{p}\mathrm{tr}(R_1(z)\widetilde{\mathbf{\Sigma}}_2) - \frac{1}{p}\mathrm{tr}(R_1(z)\mathbf{\Sigma}_p) = D_1 + D_2 + D_3, \text{ where} \\
&D_1 = \frac{1}{pn_2}\sum_{k=1}^{n_2}\left\{\mathbf{x}_{2k}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}R_1(z)\mathbf{A}\mathbf{x}_{2k} - \mathrm{tr}(R_1(z)\mathbf{A}\mathbf{\Sigma}_x\mathbf{A}^{\mathrm{T}})\right\}, \\
&D_2 = \frac{2}{pn_2}\sum_{k=1}^{n_2}\mathbf{x}_{2k}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}R_1(z)\boldsymbol{\epsilon}_{2k}, \\
&D_3 = \frac{1}{pn_2}\sum_{k=1}^{n_2}\{\boldsymbol{\epsilon}_{2k}^{\mathrm{T}}R_1(z)\boldsymbol{\epsilon}_{2k} - \mathrm{tr}(R_1(z)\mathbf{\Sigma}_\epsilon)\}.
\end{aligned}$$

Consider D_1 first. Using assumption (F1), and defining $g_i = \mathbf{x}_{2k}^{*\mathrm{T}}\mathbf{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}R_1(z)\mathbf{A}\mathbf{\Sigma}_x^{1/2}\mathbf{x}_{2k}^* - \mathrm{tr}(R_1(z)\mathbf{A}\mathbf{\Sigma}_x\mathbf{A}^{\mathrm{T}})$, for $2 \le q \le 6$ and $z \in \mathbb{C}^+$, we have by Lemma 2,

$$\begin{aligned}
E(|g_k|^q|\mathbf{Y}_1) &\le K_q(E^{q/2}|x_{2k}^*|^4\mathrm{tr}^{q/2}(\mathbf{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}R_1(z)\mathbf{A}\mathbf{\Sigma}_x\mathbf{A}^{\mathrm{T}}\overline{R_1(z)}\mathbf{A}\mathbf{\Sigma}_x^{1/2}) \\
&\quad + E|x_{2k}^*|^{2q}\mathrm{tr}(\mathbf{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}R_1(z)\mathbf{A}\mathbf{\Sigma}_x\mathbf{A}^{\mathrm{T}}\overline{R_1(z)}\mathbf{A}\mathbf{\Sigma}_x^{1/2})^{q/2}) \\
&\le K_q\big\|\mathbf{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}R_1(z)\mathbf{A}\mathbf{\Sigma}_x\mathbf{A}^{\mathrm{T}}\overline{R_1(z)}\mathbf{A}\mathbf{\Sigma}_x^{1/2}\big\|^{q/2}(E^{q/2}|x_{2k}^*|^4r^{q/2} + rE|x_{2k}^*|^{2q}) \\
&\le K_q\big\|\mathbf{\Sigma}_x\big\|^q\big\|\mathbf{A}\big\|^{2q}\mathrm{Im}^{-q}(z)(E^{q/2}|x_{2k}^*|^4r^{q/2} + rE|x_{2k}^*|^{2q}) = O(p^q),
\end{aligned}$$

where we have used $\|\mathbf{A}\| = O(p^{1/2})$ implied in assumption (F2). Using the fact that $E(g_k|\mathbf{Y}_1) = 0$ and $g_i|\mathbf{Y}_1$ is independent of $g_j|\mathbf{Y}_1$ for $i \ne j$ by assumption (F1), similar to the expansion in (7.1), we have

$$E(|D_1|^6) = \frac{1}{p^6n_2^6}\sum_{i_1\cdots i_6=1}^{n_2} E(E(g_{i_1}\bar{g}_{i_2}g_{i_3}\bar{g}_{i_4}g_{i_5}\bar{g}_{i_6}|\mathbf{Y}_1)) = O(n_2^{-3}),$$

so that $D_1 \overset{a.s.}{\to} 0$ by the Borel-Cantelli lemma, as $\sum_{n\ge1} n_2^{-3} < \infty$ by assumption.

Now consider D_2. Define $g_k = \mathbf{x}_{2k}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}R_1(z)\boldsymbol{\epsilon}_{2k}$. We have for $0 < q \le 12$ and $z \in \mathbb{C}^+$,

$$E(|g_k|^q|\mathbf{Y}_1) \le E\big\|\mathbf{x}_{2k}^*\big\|^q\big\|\mathbf{\Sigma}_x\big\|^{q/2}\big\|\mathbf{A}\big\|^qE\big\|\boldsymbol{\xi}_{2k}\big\|^q\big\|\mathbf{\Sigma}_\epsilon\big\|^{q/2}\mathrm{Im}^{-q}(z) = O(p^q),$$

by assumptions (F1) and (F2), and lemma 3. Hence, using the fact that $E(g_k|\mathbf{Y}_1) = 0$ and $g_i|\mathbf{Y}_1$ is independent of $g_j|\mathbf{Y}_1$ for $i \ne j$, an expansion similar to (7.1) leads again to

$$E(|D_2|^6) = O(n_2^{-3}),$$

so that by the Borel-Cantelli lemma, we have $D_2 \overset{a.s.}{\to} 0$.

Finally, if we define $g_k = \boldsymbol{\xi}_{2k}^{\mathrm{T}}\mathbf{\Sigma}_\epsilon^{1/2}R_1(z)\mathbf{\Sigma}_\epsilon^{1/2}\boldsymbol{\xi}_{2k} - \mathrm{tr}(R_1(z)\mathbf{\Sigma}_\epsilon)$, then following exactly the same arguments as in the proof of Theorem 1, we can conclude

$$E(|D_3|^6) = O(n_2^{-3}),$$

so that through the Borel-Cantelli lemma we have $D_3 \overset{a.s.}{\to} 0$.

Hence, we have proved

$$\frac{1}{p}\mathrm{tr}[R_1(z)\widetilde{\mathbf{\Sigma}}_2] - \frac{1}{p}\mathrm{tr}[R_1(z)\mathbf{\Sigma}_p] \overset{a.s.}{\to} 0.$$

Then, using equation (2.5) of Silverstein and Bai (1995), we conclude the almost sure convergence of the inverse Stieltjes transform on all points of continuity x, that is,

$$\frac{1}{p}\sum_{i=1}^{p}\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}\mathbf{1}_{\{\lambda_{1i}\leq x\}}-\frac{1}{p}\sum_{i=1}^{p}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}\mathbf{1}_{\{\lambda_{1i}\leq x\}}\overset{a.s.}{\to}0. \quad \square$$

Proof of Lemma 1. We first assume (A1)' and (A2)'. For $j=1,\ldots,n_2$ and $i=1,\ldots,p$, define

$$g_{ij}=\frac{\mathbf{z}_{2j}^{\mathrm{T}}\mathbf{\Sigma}_p^{1/2}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p^{1/2}\mathbf{z}_{2j}-\mathrm{tr}(\mathbf{\Sigma}_p^{1/2}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p^{1/2})}{\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}}.$$

Then for $2\leq q\leq 10$, using Lemma 2 and assumption (A1)',

$$\begin{aligned}E(|g_{ij}|^q|\mathbf{Y}_1)&\leq(\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i})^{-q}K_q\left(E^{q/2}|z_{2j}|^4\mathrm{tr}^q(\mathbf{\Sigma}_p^{1/2}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p^{1/2})+E|z_{2j}|^{2q}\mathrm{tr}(\mathbf{\Sigma}_p^{1/2}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p^{1/2})^q\right)\\&=K_q(E^{q/2}|z_{2j}|^4+E|z_{2j}|^{2q})=O(1).\end{aligned}$$

Also, $E(g_{ij}|\mathbf{Y}_1)=0$ and $g_{i_1j_1}|\mathbf{Y}_1$ is independent of $g_{i_2j_2}|\mathbf{Y}_1$ whenever $j_1\neq j_2$. Hence,

$$\begin{aligned}&E\left\{\max_{1\leq i\leq p}\left|\frac{\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2\mathbf{p}_{1i}-\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}}{\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i}}\right|\right\}^{10}=E\left\{\max_{1\leq i\leq p}\left|\frac{1}{n_2}\sum_{j=1}^{n_2}g_{ij}\right|^{10}\right\}\\&\leq\sum_{i=1}^{p}\frac{1}{n_2^{10}}\sum_{j_1\cdots j_{10}=1}^{n_2}E(E\{g_{ij_1}\cdots g_{ij_{10}}|\mathbf{Y}_1\})\\&\leq E\Bigg\{\sum_{i=1}^{p}\frac{1}{n_2^{10}}\Bigg(\sum_{j=1}^{n_2}E(g_{ij}^{10}|\mathbf{Y}_1)+\sum_{\substack{q_1,q_2\neq1\\q_1+q_2=10}}\sum_{j_1\neq j_2}E(g_{ij_1}^{q_1}|\mathbf{Y}_1)E(g_{ij_2}^{q_2}|\mathbf{Y}_1)\\&+\sum_{\substack{q_1,q_2,q_3\neq1\\q_1+q_2+q_3=10}}\sum_{j_1\neq j_2\neq j_3}\prod_{k=1}^{3}E(g_{ij_k}^{q_k}|\mathbf{Y}_1)+\sum_{\substack{q_1,\ldots,q_4\neq1\\q_1+\cdots+q_4=10}}\sum_{j_1\neq j_2\neq j_3\neq j_4}\prod_{k=1}^{4}E(g_{ij_k}^{q_k}|\mathbf{Y}_1)\\&+\sum_{j_1\neq\cdots\neq j_5}\prod_{k=1}^{5}E(g_{ij_k}^{2}|\mathbf{Y}_1)\Bigg)\Bigg\}\\&=O(pn_2^{-10}[n_2+4\mathrm{P}_2^{n_2}+4\mathrm{P}_3^{n_2}+2\mathrm{P}_4^{n_2}+\mathrm{P}_5^{n_2}])\\&=O(pn_2^{-5}),\end{aligned}\tag{7.2}$$

where $\mathrm{P}_r^{n_2}=n_2(n_2-1)\cdots(n_2-r+1)$. Since $\sum_{n\geq1}pn_2^{-5}<\infty$ by assumption, the Borel-Cantelli lemma completes the proof of this part.

Now consider data from a factor model, with assumption (F1)' satisfied. We can decompose

$$\frac{\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i}-\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}} = D_1+D_2+D_3, \quad \text{where}$$

$$D_1=\frac{1}{n_2}\sum_{j=1}^{n_2} g_{ij}, \quad g_{ij}=\frac{\mathbf{x}_{2j}^{*\mathrm{T}}\boldsymbol{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{A}\boldsymbol{\Sigma}_x^{1/2}\mathbf{x}_{2j}^{*}-\mathrm{tr}(\boldsymbol{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{A}\boldsymbol{\Sigma}_x^{1/2})}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}},$$

$$D_2=\frac{1}{n_2}\sum_{j=1}^{n_2} d_{ij}, \quad d_{ij}=\frac{\boldsymbol{\xi}_{2j}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\boldsymbol{\xi}_{2j}-\mathrm{tr}(\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2})}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}},$$

$$D_3=\frac{2}{n_2}\sum_{j=1}^{n_2} h_{ij}, \quad h_{ij}=\frac{\mathbf{x}_{2j}^{*\mathrm{T}}\boldsymbol{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\boldsymbol{\xi}_{2j}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}.$$

Using Lemma 2, for $2\le q\le 10$,

$$\begin{aligned}
E(|g_{ij}|^q|\mathbf{Y}_1) &\le (\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})^{-q}K_q(\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{A}\boldsymbol{\Sigma}_x\mathbf{A}^{\mathrm{T}}\mathbf{p}_{1i})^q(E^{q/2}|x_{2j}^*|^4+E|x_{2j}^*|^{2q})\le K_q(E^{q/2}|x_{2j}^*|^4+E|x_{2j}^*|^{2q})\\
&=O(1);\\
E(|d_{ij}|^q|\mathbf{Y}_1) &\le (\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})^{-q}K_q(\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon\mathbf{p}_{1i})^q(E^{q/2}|\xi_{2j}|^4+E|\xi_{2j}|^{2q})\le K_q(E^{q/2}|\xi_{2j}|^4+E|\xi_{2j}|^{2q})\\
&=O(1).
\end{aligned}$$

For $4\le q\le 10$,

$$\begin{aligned}
E(|h_{ij}|^q|\mathbf{Y}_1) &= E\left(\left(\frac{\mathbf{x}_{2j}^{*\mathrm{T}}\boldsymbol{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{A}\boldsymbol{\Sigma}_x^{1/2}\mathbf{x}_{2j}^{*}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}\right)^{q/2}\Bigg|\mathbf{Y}_1\right) E\left(\left(\frac{\boldsymbol{\xi}_{2j}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\boldsymbol{\xi}_{2j}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}\right)^{q/2}\Bigg|\mathbf{Y}_1\right),\\
\text{with } & E\left(\left(\frac{\mathbf{x}_{2j}^{*\mathrm{T}}\boldsymbol{\Sigma}_x^{1/2}\mathbf{A}^{\mathrm{T}}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{A}\boldsymbol{\Sigma}_x^{1/2}\mathbf{x}_{2j}^{*}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}\right)^{q/2}\Bigg|\mathbf{Y}_1\right)\le C_{q/2}(E(|g_{ij}|^{q/2}|\mathbf{Y}_1)+1)=O(1),\\
& E\left(\left(\frac{\boldsymbol{\xi}_{2j}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{p}_{1i}\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\boldsymbol{\xi}_{2j}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}\right)^{q/2}\Bigg|\mathbf{Y}_1\right)\le C_{q/2}(E(|d_{ij}|^{q/2}|\mathbf{Y}_1)+1)=O(1),
\end{aligned}$$

where $C_{q/2}$ is a constant. Hence, $E(|h_{ij}|^q|\mathbf{Y}_1)=O(1)$ for $q\ge 4$. For $2\le q<4$, it is easy to see that $E(|h_{ij}|^q|\mathbf{Y}_1)\le E^{1/2}(|h_{ij}|^{2q}|\mathbf{Y}_1)=O(1)$, and hence $E(|h_{ij}|^q|\mathbf{Y}_1)=O(1)$ for $2\le q\le 10$. Then since $E(g_{ij}|\mathbf{Y}_1)=0$ and $g_{i_1j_1}|\mathbf{Y}_1$ is independent of $g_{i_2j_2}|\mathbf{Y}_1$ whenever $j_1\ne j_2$, an expansion exactly like (7.2) will show that

$$E\left(\left|\max_{1\le i\le p}|D_1|^{10}\right|\right)=O(pn_2^{-5}),$$

showing that $\max_{1\le i\le p}|D_1|\overset{a.s.}{\to}0$, since $\sum_{n\ge1}pn_2^{-5}<\infty$ by assumption. Using the same expansion (7.2) and the arguments above, we can also show that $\max_{1\le i\le p}|D_2|,\ \max_{1\le i\le p}|D_3|\overset{a.s.}{\to}0$. This completes the proof of the lemma. □

Before proving Theorem 5, we state and prove a lemma first.

Lemma 5. *Let assumptions (A1) to (A4) be satisfied. Then assuming* $\mathbf{\Sigma}_p \neq \sigma^2\mathbf{I}$*,*

$$\frac{\big\|\widehat{\mathbf{\Sigma}}_{\text{Ideal},m} - \mathbf{\Sigma}_p\big\|_F^2}{\big\|\widehat{\mathbf{\Sigma}}_{\text{Ideal}} - \mathbf{\Sigma}_p\big\|_F^2} \overset{a.s.}{\rightarrow} 1, \quad \frac{\log\det(\widehat{\mathbf{\Sigma}}_{\text{Ideal},m}\mathbf{\Sigma}_p^{-1})}{\log\det(\widehat{\mathbf{\Sigma}}_{\text{Ideal}}\mathbf{\Sigma}_p^{-1})} \overset{a.s.}{\rightarrow} 1.$$

Proof of Lemma 5. Consider the Frobenius loss first. For λ_i the ith largest eigenvalue of $\mathbf{S}_n$ with corresponding eigenvector $\mathbf{p}_i$,

$$\begin{aligned}
p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{\text{Ideal}} - \mathbf{\Sigma}_p\big\|_F^2 &= p^{-1}\text{tr}(\text{diag}(\mathbf{P}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}) - \mathbf{P}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P})^2 \\
&= p^{-1}\sum_{i=1}^{n}(\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i)^2 - 2p^{-1}\text{tr}(\mathbf{P}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P}\text{diag}(\mathbf{P}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{P})) + p^{-1}\text{tr}(\mathbf{\Sigma}_p^2) \\
&= p^{-1}\text{tr}(\mathbf{\Sigma}_p^2) - p^{-1}\sum_{i=1}^{p}(\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i)^2,
\end{aligned}$$

so that similarly, we have $p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{\text{Ideal},m} - \mathbf{\Sigma}_p\big\|_F^2 = p^{-1}\text{tr}(\mathbf{\Sigma}_p^2) - p^{-1}\sum_{i=1}^{p}(\mathbf{p}_{1i}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i})^2$. But with assumption (A3), we have

$$p^{-1}\text{tr}(\mathbf{\Sigma}_p^2) = p^{-1}\sum_{i=1}^{p}\tau_{n,i}^2 \overset{a.s.}{\rightarrow} \int \tau^2 dH(\tau).$$

Moreover, with the convergence result in Theorem 4 of Ledoit and Péché (2011) indicating that (see equation (2.7) as well) for all $x \in \mathbb{R}$, we have

$$\Delta_p(x) = p^{-1}\sum_{i=1}^{p}\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i\mathbf{1}_{\{\lambda_i \leq x\}} \overset{a.s.}{\rightarrow} \Delta(x) = \int_{-\infty}^{x}\delta(\lambda)dF(\lambda),$$

where $F(\cdot)$ is such that

$$F_p(\lambda) = p^{-1}\sum_{i=1}^{p}\mathbf{1}_{\{\lambda_i \leq \lambda\}} \overset{a.s.}{\rightarrow} F(\lambda),$$

we have for any continuous function $g(\cdot)$ over the positive real line,

$$p^{-1}\sum_{i=1}^{p}g(\mathbf{p}_i^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_i)\mathbf{1}_{\{\lambda_i \leq x\}} \overset{a.s.}{\rightarrow} \int_{-\infty}^{x}g(\delta(\lambda))dF(\lambda). \tag{7.3}$$

With this, it is easy to see that then

$$p^{-1}\big\|\widehat{\mathbf{\Sigma}}_{\text{Ideal}} - \mathbf{\Sigma}_p\big\|_F^2 \overset{a.s.}{\rightarrow} \int \tau^2 dH(\tau) - \int \delta^2(\lambda)dF(\lambda),$$

which is nonzero in general, except when $\mathbf{\Sigma}_p = \sigma^2\mathbf{I}$. At the same time, we have (see equation (2.9) and the descriptions therein as well)

$$p^{-1} \sum_{i=1}^{p} \mathbf{p}_{1i}^{\mathrm{T}} \mathbf{\Sigma}_p \mathbf{p}_{1i} \mathbf{1}_{\{\lambda_{1i} \le x\}} \xrightarrow{a.s.} \int_{-\infty}^{x} \delta_1(\lambda) dF_1(\lambda),$$

where $F_1(\cdot)$ is such that

$$F_{1p}(\lambda) = p^{-1} \sum_{i=1}^{p} \mathbf{1}_{\{\lambda_{1i} \le \lambda\}} \xrightarrow{a.s.} F_1(\lambda).$$

But since $m/n = n_1/n \to 1$, we have $p/n_1, p/n$ both going to the same limit $c_1 = c > 0$. Theorem 4.1 of Bai and Silverstein (2010) tells us then both F_p and F_{1p} converges to the same limit almost surely under assumptions (A1) to (A4). Hence $F = F_1$ almost surely,implying $\delta_1(\cdot) = \delta(\cdot)$ almost surely (See Remark 1 in section 2.2 as well). This immediately implies that, similar to (7.3),

$$p^{-1} \sum_{i=1}^{p} (\mathbf{p}_{1i}^{\mathrm{T}} \mathbf{\Sigma}_p \mathbf{p}_{1i})^2 \mathbf{1}_{\{\lambda_{1i} \le x\}} \xrightarrow{a.s.} \int_{-\infty}^{x} \delta_1^2(\lambda) dF_1(\lambda) = \int_{-\infty}^{x} \delta^2(\lambda) dF(\lambda),$$

and hence

$$p^{-1} \big\| \widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m} - \mathbf{\Sigma}_p \big\|_F^2 \xrightarrow{a.s.} \int \tau^2 dH(\tau) - \int \delta_1^2(\lambda) dF_1(\lambda) = \int \tau^2 dH(\tau) - \int \delta^2(\lambda) dF(\lambda).$$

We immediately have

$$\frac{\big\| \widehat{\mathbf{\Sigma}}_{\mathrm{Ideal},m} - \mathbf{\Sigma}_p \big\|_F^2}{\big\| \widehat{\mathbf{\Sigma}}_{\mathrm{Ideal}} - \mathbf{\Sigma}_p \big\|_F^2} \xrightarrow{a.s.} 1,$$

which completes the proof for the Frobenius loss.

For the remaining part, note that it is not difficult to show the inverse Stein's loss

$$p^{-1} SL(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_{\mathrm{Ideal}}) = p^{-1} \log\det(\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal}} \mathbf{\Sigma}_p^{-1}) = p^{-1} \sum_{i=1}^{p} \log(\mathbf{p}_i^{\mathrm{T}} \mathbf{\Sigma}_p \mathbf{p}_i) - p^{-1} \sum_{i=1}^{p} \log(\tau_{n,i}).$$

Similar arguments as before show that

$$p^{-1} \sum_{i=1}^{p} \log(\mathbf{p}_i^{\mathrm{T}} \mathbf{\Sigma}_p \mathbf{p}_i) \xrightarrow{a.s.} \int \log(\delta(\lambda)) dF(\lambda),$$
$$p^{-1} \sum_{i=1}^{p} \log(\tau_{n,i}) \xrightarrow{a.s.} \int \log(\tau) dH(\tau).$$

Hence

$$p^{-1} SL(\mathbf{\Sigma}_p, \widehat{\mathbf{\Sigma}}_{\mathrm{Ideal}}) = p^{-1} \log\det(\widehat{\mathbf{\Sigma}}_{\mathrm{Ideal}} \mathbf{\Sigma}_p^{-1}) \xrightarrow{a.s.} \int \log(\delta(\lambda)) dF(\lambda) - \int \log(\tau) dH(\tau),$$

which is nonzero in general except when $\mathbf{\Sigma}_p = \sigma^2 \mathbf{I}$. Moreover, similar arguments as in the case for the

Frobenius loss show that

$$p^{-1}\sum_{i=1}^{p}\log(\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})\mathbf{1}_{\{\lambda_{1i}\le x\}} \overset{a.s.}{\to} \int_{-\infty}^{x}\log(\delta_1(\lambda))dF_1(\lambda) = \int_{-\infty}^{x}\log(\delta(\lambda))dF(\lambda).$$

Hence,

$$\begin{aligned} p^{-1}SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal},m}) = p^{-1}\log\det(\widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal},m}\boldsymbol{\Sigma}_p^{-1}) &\overset{a.s.}{\to} \int\log(\delta_1(\lambda))dF_1(\lambda) - \int\log(\tau)dH(\tau) \\ &= \int\log(\delta(\lambda))dF(\lambda) - \int\log(\tau)dH(\tau). \end{aligned}$$

This completes the proof of the lemma. □

Proof of Theorem 5. We first assume (A1)', (A2)', (A3) and (A4). Since we have

$$\begin{aligned} p^{-1}\big\|\widehat{\boldsymbol{\Sigma}}_m - \boldsymbol{\Sigma}_p\big\|_F^2 &= p^{-1}\big\|\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{P}_1) - \mathbf{P}_1^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{P}_1\big\|_F^2 \\ &= p^{-1}\sum_{i=1}^{p}(\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i} - \mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})^2 + p^{-1}\big\|\widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal},m} - \boldsymbol{\Sigma}_p\big\|_F^2, \end{aligned}$$

we can express the efficiency loss in (4.2) with respect to the Frobenius loss as

$$EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m) = 1 - \left(\frac{p^{-1}\sum_{i=1}^{p}(\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i} - \mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})^2}{p^{-1}\big\|\widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal}} - \boldsymbol{\Sigma}_p\big\|_F^2} + \frac{p^{-1}\big\|\widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal},m} - \boldsymbol{\Sigma}_p\big\|_F^2}{p^{-1}\big\|\widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal}} - \boldsymbol{\Sigma}_p\big\|_F^2}\right)^{-1}.$$

With $m/n \to 1$, we have $p/m \to c > 0$, same as p/n does. The conclusion from lemma 5 that the ratio of loss for the ideal estimators being almost surely 1 then implies that $EL(\widehat{\boldsymbol{\Sigma}}_m - \boldsymbol{\Sigma}_p) \overset{a.s.}{\to} 0$, if we can show that $p^{-1}\sum_{i=1}^{p}(\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i} - \mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})^2 \overset{a.s.}{\to} 0$. But together with assumption $\sum_{n\ge1} pn_2^{-5} < \infty$, we can apply Lemma 1, so that

$$p^{-1}\sum_{i=1}^{p}(\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i} - \mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})^2 \le \left(\max_{1\le i\le p}\left|\frac{\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i} - \mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}\right|\right)^2 \max_{1\le i\le p}(\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})^2 \overset{a.s.}{\to} 0,$$

since $\max_{1\le i\le p}(\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i})^2 \le \big\|\boldsymbol{\Sigma}_p\big\|^2 = O(1)$ by assumption (A2)'. Hence, $EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m) \overset{a.s.}{\to} 0$ in this case.

Now consider the inverse Stein's loss. We have

$$\begin{aligned} SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m) &= \mathrm{tr}(\mathbf{P}_1^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{P}_1\mathrm{diag}^{-1}(\mathbf{P}_1^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{P}_1)) - \log\det(\mathbf{P}_1^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{P}_1\mathrm{diag}^{-1}(\mathbf{P}_1^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{P}_1)) - p \\ &= \sum_{i=1}^{p}\frac{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}{\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i}} - p + \sum_{i=1}^{p}\log\left(\frac{\mathbf{p}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}\right) + SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal},m}) \\ &= \sum_{i=1}^{p}h_1(g_i) + \sum_{i=1}^{p}h_2(g_i) + SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal},m}), \end{aligned}$$

where we define g_i for $i = 1, \ldots, p$, and the functions h_1 and h_2, which are differentiable on $(-1, \infty)$, to be

$$g_i = \frac{\mathbf{p}_{1i}\widetilde{\boldsymbol{\Sigma}}_2\mathbf{p}_{1i} - \mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}{\mathbf{p}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i}}, \quad h_1(x) = \frac{1}{1+x} - 1, \quad h_2(x) = \log(1+x).$$

With $SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\text{Ideal},m})/SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\text{Ideal}}) \overset{a.s.}{\to} 1$ by lemma 5,

$$\begin{aligned} EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m) &= 1 - \left(\frac{p^{-1}\sum_{i=1}^p h_1(g_i)}{p^{-1}SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\text{Ideal}})} + \frac{p^{-1}\sum_{i=1}^p h_2(g_1)}{p^{-1}SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\text{Ideal}})} + \frac{p^{-1}SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\text{Ideal},m})}{p^{-1}SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\text{Ideal}})}\right)^{-1} \\ &\overset{a.s.}{\to} 0, \end{aligned}$$

if we can show further that $p^{-1}\sum_{i=1}^p h_j(g_i) \overset{a.s.}{\to} 0$ for $j = 1, 2$. To this end, by Lemma 1, g_i is almost surely 0 for all i. Hence, in particular, for n large enough, we have $|g_i| \leq B < 1$ almost surely, so that $h_j(g_i)$ is well defined for all i. This implies that for $j = 1, 2$,

$$p^{-1}\sum_{i=1}^p h_j(g_i) \leq \max_{1\leq i\leq p} h_j(g_i) = \max_{1\leq i\leq p} |g_i||h_j'(\eta_i)| \overset{a.s.}{\to} 0,$$

where η_i lies almost surely in $[-B, B]$ for all i for large enough n, so that the $h_j'(\eta_i)$'s are bounded almost surely for all i. This completes the proof of the theorem. □

Proof of Theorem 6. For the Frobenius loss,

$$\big\|\widehat{\boldsymbol{\Sigma}}_{m,M} - \boldsymbol{\Sigma}_p\big\|_F^2 = \Big\|\frac{1}{M}\sum_{i=1}^M(\widehat{\boldsymbol{\Sigma}}_m^{(i)} - \boldsymbol{\Sigma}_p)\Big\|_F^2 \leq \Big(\frac{1}{M}\sum_{i=1}^M\big\|\widehat{\boldsymbol{\Sigma}}_m^{(i)} - \boldsymbol{\Sigma}_p\big\|_F\Big)^2 \leq \frac{1}{M}\sum_{i=1}^M\big\|\widehat{\boldsymbol{\Sigma}}_m^{(i)} - \boldsymbol{\Sigma}_p\big\|_F^2,$$

so that

$$EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{m,M}) \leq 1 - \frac{\big\|\widehat{\boldsymbol{\Sigma}}_{\text{Ideal}} - \boldsymbol{\Sigma}_p\big\|_F^2}{\frac{1}{M}\sum_{i=1}^M\big\|\widehat{\boldsymbol{\Sigma}}_m^{(i)} - \boldsymbol{\Sigma}_p\big\|_F^2} = 1 - \frac{1}{\frac{1}{M}\sum_{i=1}^M\frac{1}{1-EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m^{(i)})}} \overset{a.s.}{\to} 0,$$

since $EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m^{(i)}) \overset{a.s.}{\to} 0$ by Theorem 5. This completes the proof for the Frobenius loss.

For the inverse Stein's loss, note that for any estimator $\widehat{\boldsymbol{\Sigma}}$,

$$SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}) = \operatorname{tr}(\boldsymbol{\Sigma}_p\widehat{\boldsymbol{\Sigma}}^{-1}) - p - \log\det(\boldsymbol{\Sigma}_p\widehat{\boldsymbol{\Sigma}}^{-1}) = \operatorname{tr}\Big((\boldsymbol{\Sigma}_p - \widehat{\boldsymbol{\Sigma}})\widehat{\boldsymbol{\Sigma}}^{-1}\Big) + \log\det(\widehat{\boldsymbol{\Sigma}}\boldsymbol{\Sigma}_p^{-1}).$$

Hence

$$
\begin{aligned}
SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{m,M}) &= \frac{1}{M}\sum_{i=1}^{M} \mathrm{tr}\Big((\boldsymbol{\Sigma}_p - \widehat{\boldsymbol{\Sigma}}_m^{(i)})\widehat{\boldsymbol{\Sigma}}_{m,M}^{-1}\Big) + \log\det\Big(\frac{1}{M}\sum_{i=1}^{M}\widehat{\boldsymbol{\Sigma}}_m^{(i)}\boldsymbol{\Sigma}_p^{-1}\Big) \\
&\le \frac{1}{M}\sum_{i=1}^{M} \mathrm{tr}\Big((\boldsymbol{\Sigma}_p - \widehat{\boldsymbol{\Sigma}}_m^{(i)})\widehat{\boldsymbol{\Sigma}}_{m,M}^{-1}\Big) + \frac{1}{M}\sum_{i=1}^{M}\log\det(\widehat{\boldsymbol{\Sigma}}_m^{(i)}\boldsymbol{\Sigma}_p^{-1}) \\
&= \frac{1}{M}\sum_{i=1}^{M} \mathrm{tr}\Big((\boldsymbol{\Sigma}_p - \widehat{\boldsymbol{\Sigma}}_m^{(i)})\big(\widehat{\boldsymbol{\Sigma}}_{m,M}^{-1} - (\widehat{\boldsymbol{\Sigma}}_m^{(i)})^{-1}\big)\Big) + \frac{1}{M}\sum_{i=1}^{M} SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m^{(i)}),
\end{aligned}
$$

where the second line comes from the convexity of $\log\det(\mathbf{X})$ where $\mathbf{X}$ is positive semi-definite. Then

$$
\begin{aligned}
EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{m,M}) \le 1 - &\left(\frac{\frac{1}{pM}\sum_{i=1}^{M}\mathrm{tr}\Big((\boldsymbol{\Sigma}_p - \widehat{\boldsymbol{\Sigma}}_m^{(i)})\big(\widehat{\boldsymbol{\Sigma}}_{m,M}^{-1} - (\widehat{\boldsymbol{\Sigma}}_m^{(i)})^{-1}\big)\Big)}{p^{-1}SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal}})} + \frac{1}{M}\sum_{i=1}^{M}\frac{1}{1 - EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m^{(i)})}\right)^{-1} \\
&\overset{a.s.}{\to} 0,
\end{aligned}
$$

if we can show further that $\frac{1}{pM}\sum_{i=1}^{M}\mathrm{tr}\Big((\boldsymbol{\Sigma}_p - \widehat{\boldsymbol{\Sigma}}_m^{(i)})\big(\widehat{\boldsymbol{\Sigma}}_{m,M}^{-1} - (\widehat{\boldsymbol{\Sigma}}_m^{(i)})^{-1}\big)\Big) \overset{a.s.}{\to} 0$, since by Theorem 5 we have $EL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_m^{(i)}) \overset{a.s.}{\to} 0$ for each i, and that in the proof of Lemma 5 it is proved that $p^{-1}SL(\boldsymbol{\Sigma}_p, \widehat{\boldsymbol{\Sigma}}_{\mathrm{Ideal}})$ converges almost surely to a nonzero value when $\boldsymbol{\Sigma}_p \neq \sigma^2\mathbf{I}$. For this, write the term as $D_1 + D_2$, where

$$
D_1 = \frac{1}{pM}\sum_{i=1}^{M}\mathrm{tr}\big((\widehat{\boldsymbol{\Sigma}}_m^{(i)} - \boldsymbol{\Sigma}_p)(\widehat{\boldsymbol{\Sigma}}_m^{(i)})^{-1}\big), \quad D_2 = \frac{1}{pM}\sum_{i=1}^{M}\mathrm{tr}\big((\boldsymbol{\Sigma}_p - \widehat{\boldsymbol{\Sigma}}_m^{(i)})\widehat{\boldsymbol{\Sigma}}_{m,M}^{-1}\big).
$$

For D_1, writing $\mathbf{P}_{1i} = (\mathbf{p}_{1i,1}, \ldots, \mathbf{p}_{1i,p})$, we can use Lemma 1 to deduce that

$$
\begin{aligned}
D_1 &= \frac{1}{pM}\sum_{i=1}^{M}\mathrm{tr}\big((\mathrm{diag}(\mathbf{P}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2^{(i)}\mathbf{P}_{1i}) - \mathbf{P}_{1i}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{P}_{1i})\mathrm{diag}^{-1}(\mathbf{P}_{1i}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2^{(i)}\mathbf{P}_{1i})\big) \\
&= \frac{1}{pM}\sum_{i=1}^{M}\sum_{j=1}^{p}\frac{\mathbf{p}_{1i,j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2^{(i)}\mathbf{p}_{1i,j} - \mathbf{p}_{1i,j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i,j}}{\mathbf{p}_{1i,j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2^{(i)}\mathbf{p}_{1i,j}} \\
&\le \frac{1}{M}\sum_{i=1}^{M}\max_{1\le j\le p}\left|\frac{\mathbf{p}_{1i,j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2^{(i)}\mathbf{p}_{1i,j} - \mathbf{p}_{1i,j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i,j}}{\mathbf{p}_{1i,j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i,j}}\right| \cdot \left(1 - \max_{1\le j\le p}\left|\frac{\mathbf{p}_{1i,j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2^{(i)}\mathbf{p}_{1i,j} - \mathbf{p}_{1i,j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i,j}}{\mathbf{p}_{1i,j}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i,j}}\right|\right)^{-1} \overset{a.s.}{\to} 0.
\end{aligned}
$$

For D_2, define $\widehat{\boldsymbol{\Sigma}}_{\mathrm{CRC}}^{(i)} = \mathbf{P}_{1i}\Big(\frac{1}{M}\sum_{j=1}^{M}\mathrm{diag}(\mathbf{P}_{1j}^{\mathrm{T}}\widetilde{\boldsymbol{\Sigma}}_2^{(j)}\mathbf{P}_{1j})\Big)\mathbf{P}_{1i}$, and $\mathbf{B}_i = (\widehat{\boldsymbol{\Sigma}}_{\mathrm{CRC}}^{(i)} - \widehat{\boldsymbol{\Sigma}}_{m,M})(\widehat{\boldsymbol{\Sigma}}_{\mathrm{CRC}}^{(i)})^{-1}$. Then

$$
\widehat{\boldsymbol{\Sigma}}_{m,M}^{-1} = (\widehat{\boldsymbol{\Sigma}}_{\mathrm{CRC}}^{(i)})^{-1}(\mathbf{I} - \mathbf{B}_i)^{-1} = (\widehat{\boldsymbol{\Sigma}}_{\mathrm{CRC}}^{(i)})^{-1} + (\widehat{\boldsymbol{\Sigma}}_{\mathrm{CRC}}^{(i)})^{-1}\sum_{k\ge 1}\mathbf{B}_i^k. \tag{7.4}
$$

Suppose $\max_{1\le i,j\le M}\big\|\mathbf{P}_{1i} - \mathbf{P}_{1j}\big\| \overset{a.s.}{\to} 0$, to be proved at the end of this proof. Then the above Neumann's series expansion is valid, since then for large enough n, p with $p/n \to c > 0$,

$$\|\mathbf{B}_i\| \leq \frac{1}{M}\left(2\left\|\sum_{j=1}^{M}(\mathbf{P}_{1i}-\mathbf{P}_{1j})\mathrm{diag}(\mathbf{P}_{1j}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(j)}\mathbf{P}_{1j})\mathbf{P}_{1i}^{\mathrm{T}}\right\| + \left\|\sum_{j=1}^{M}(\mathbf{P}_{1i}-\mathbf{P}_{1j})\mathrm{diag}(\mathbf{P}_{1j}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(j)}\mathbf{P}_{1j})(\mathbf{P}_{1i}-\mathbf{P}_{1j})^{\mathrm{T}}\right\|\right)$$
$$\cdot\Big(\min_{1\leq j\leq M}\min_{1\leq k\leq p}\mathbf{p}_{1j,k}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(j)}\mathbf{p}_{1j,k}\Big)^{-1}$$
$$\leq \max_{1\leq j\leq M}\|\mathbf{P}_{1i}-\mathbf{P}_{1j}\|\max_{1\leq k\leq p}\mathbf{p}_{1j,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j,k}\Big(2+\|\mathbf{P}_{1i}-\mathbf{P}_{1j}\|\Big)\cdot\Big(\min_{1\leq j\leq M}\min_{1\leq k\leq p}\mathbf{p}_{1j,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j,k}\Big)^{-1}$$
$$\leq \max_{1\leq i,j\leq M}\|\mathbf{P}_{1i}-\mathbf{P}_{1j}\|\|\mathbf{\Sigma}_p\|(2+\|\mathbf{P}_{1i}-\mathbf{P}_{1j}\|)/\lambda_{\min}(\mathbf{\Sigma}_p) \overset{a.s.}{\to} 0, \tag{7.5}$$

where the second last line follows from Lemma 1 when n, p is large. With (7.4), $D_2 = D_3 + D_4$, where

$$D_3 = \frac{1}{pM}\sum_{i=1}^{M}\mathrm{tr}\Big((\mathbf{\Sigma}_p - \widehat{\mathbf{\Sigma}}_m^{(i)})(\widehat{\mathbf{\Sigma}}_{\mathrm{CRC}}^{(i)})^{-1}\Big), \quad D_4 = \frac{1}{pM}\sum_{i=1}^{M}\mathrm{tr}\Big((\mathbf{\Sigma}_p - \widehat{\mathbf{\Sigma}}_m^{(i)})(\widehat{\mathbf{\Sigma}}_{\mathrm{CRC}}^{(i)})^{-1}\sum_{k\geq 1}\mathbf{B}_i^k\Big).$$

Similar to the analysis of D_1, using Lemma 1, we have

$$D_3 = \frac{1}{pM}\sum_{i=1}^{M}\sum_{k=1}^{p}\frac{\mathbf{p}_{1i,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i,k} - \mathbf{p}_{1i,k}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(i)}\mathbf{p}_{1i,k}}{\frac{1}{M}\sum_{j=1}^{M}\mathbf{p}_{1j,k}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(j)}\mathbf{p}_{1j,k}}$$
$$\leq \max_{1\leq i\leq M}\max_{1\leq k\leq p}\left|\frac{\mathbf{p}_{1i,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i,k} - \mathbf{p}_{1i,k}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(i)}\mathbf{p}_{1i,k}}{\mathbf{p}_{1i,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i,k}}\right|\cdot\mathbf{p}_{1i,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i,k}$$
$$\cdot\left(\min_{1\leq j\leq M}\min_{1\leq k\leq p}\left(\frac{\mathbf{p}_{1j,k}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(j)}\mathbf{p}_{1j,k} - \mathbf{p}_{1j,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j,k}}{\mathbf{p}_{1j,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j,k}}\right)\mathbf{p}_{1j,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j,k} + \mathbf{p}_{1j,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1j,k}\right)^{-1} \overset{a.s.}{\to} 0.$$

For D_4, with $\|\mathbf{B}_i\| \overset{a.s.}{\to} 0$ proved in (7.5) and using Lemma 1, we have

$$D_4 \leq \max_{1\leq i\leq M}\|\widehat{\mathbf{\Sigma}}_m^{(i)} - \mathbf{\Sigma}_p\|\cdot\lambda_{\min}^{-1}(\widehat{\mathbf{\Sigma}}_{\mathrm{CRC}}^{(i)})\cdot\sum_{k\geq 1}\|\mathbf{B}_i\|^k$$
$$\leq \frac{(\|\mathbf{\Sigma}_p\| + \max_{1\leq i\leq M}\max_{1\leq k\leq p}\mathbf{p}_{1i,k}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(i)}\mathbf{p}_{1i,k})\|\mathbf{B}_i\|}{\min_{1\leq i\leq M}\min_{1\leq k\leq p}\mathbf{p}_{1i,k}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}_2^{(i)}\mathbf{p}_{1i,k}(1-\|\mathbf{B}_i\|)} \overset{a.s.}{\to} 0.$$

Hence the proof of the Theorem is complete if we can show that $\max_{1\leq i,j\leq M}\|\mathbf{P}_{1i}-\mathbf{P}_{1j}\| \overset{a.s.}{\to} 0$. To this end, using (7.3), we have for any $1 \leq i \leq M$, with $\lambda_{1i,k}$ denoting the kth largest eigenvalue of $\widetilde{\mathbf{\Sigma}}_1^{(i)}$,

$$p^{-1}\sum_{k=1}^{p}g(\mathbf{p}_{1i,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i,k})\mathbf{1}_{\{\lambda_{1i,k}\leq x\}} \overset{a.s.}{\to} \int_{-\infty}^{x}g(\delta(\lambda))dF(\lambda),$$

since $m/n \to 1$ implies that $p/n, p/m \to c > 0$. Setting $g \equiv 1$, it is easy to see that $\lambda_{1i,k}$ is almost surely the same as $\lambda_{1j,k}$ for $1 \leq i, j \leq M$ and $1 \leq k \leq p$, as $n, p \to \infty$. These imply that for $1 \leq i \leq M$, $\mathbf{p}_{1i,k}^{\mathrm{T}}\mathbf{\Sigma}_p\mathbf{p}_{1i,k}$

is almost surely the same as $\delta(\lambda_{11,k})$, and hence for $1 \le i, j \le M$ and $1 \le k \le p$,

$$\mathbf{p}_{1i,k}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1j,k} \overset{a.s.}{\to} 0.$$

But $\mathbf{p}_{1i,k}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k}^{\mathrm{T}}\boldsymbol{\Sigma}_p\mathbf{p}_{1j,k} = (\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^{\mathrm{T}}\boldsymbol{\Sigma}_p(\mathbf{p}_{1i,k} + \mathbf{p}_{1j,k})$, hence if $\mathbf{p}_{1i,k}$ or $\mathbf{p}_{1j,k}$ are not eigenvectors of $\boldsymbol{\Sigma}_p$ asymptotically (this excludes the case $\boldsymbol{\Sigma}_p = \sigma^2\mathbf{I}$, which by assumption is not the case), then there exists a constant $a > 0$ such that

$$(\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^{\mathrm{T}}\boldsymbol{\Sigma}_p(\mathbf{p}_{1i,k} + \mathbf{p}_{1j,k}) = \big\|\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k}\big\| \cdot a\big\|\mathbf{p}_{1i,k} + \mathbf{p}_{1j,k}\big\| \overset{a.s.}{\to} 0,$$

implying that for all $1 \le i, j \le M$, $1 \le k \le p$, $\big\|\mathbf{p}_{1i,k} \pm \mathbf{p}_{1j,k}\big\| \overset{a.s.}{\to} 0$. With the correct orientation chosen, we must then have $\max_{1\le i,j\le M} \big\|\mathbf{P}_i - \mathbf{P}_j\big\| \overset{a.s.}{\to} 0$.

Finally, if both $\mathbf{p}_{1i,k}$ and $\mathbf{p}_{1j,k}$ are asymptotic eigenvectors of $\boldsymbol{\Sigma}_p$ for some $1 \le k \le p$, then assuming $\big\|\boldsymbol{\Sigma}_p\mathbf{p}_{1i,k} - \lambda\mathbf{p}_{1i,k}\big\| \overset{a.s.}{\to} 0$ and $\big\|\boldsymbol{\Sigma}_p\mathbf{p}_{1j,k} - \gamma\mathbf{p}_{1j,k}\big\| \overset{a.s.}{\to} 0$,

$$\begin{aligned}(\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^{\mathrm{T}}\boldsymbol{\Sigma}_p(\mathbf{p}_{1i,k} + \mathbf{p}_{1j,k}) &= (\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^{\mathrm{T}}(\lambda\mathbf{p}_{1i,k} + \gamma\mathbf{p}_{1j,k}) \\ &\quad + (\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^{\mathrm{T}}(\boldsymbol{\Sigma}_p\mathbf{p}_{1i,k} - \lambda\mathbf{p}_{1i,k} + \boldsymbol{\Sigma}_p\mathbf{p}_{1j,k} - \gamma\mathbf{p}_{1j,k}) \\ &= (\lambda - \gamma)(1 - \mathbf{p}_{1i,k}^{\mathrm{T}}\mathbf{p}_{1j,k}) + o(1),\end{aligned}$$

so that if $\lambda \neq \gamma$, we must have $\big\|\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k}\big\| \overset{a.s.}{\to} 0$, which is what we want. If $\lambda = \gamma$, then $\mathbf{p}_{1i,k}$ and $\mathbf{p}_{1j,k}$ are asymptotically spanning the same eigenspace, hence $\max_{1\le i,j\le M} \big\|\mathbf{P}_i - \mathbf{P}_j\big\| \overset{a.s.}{\to} 0$ still holds. This completes the proof of the theorem. □

References

Abadir, K. M., Distaso, W., and Žikeš, F. (2014). Design-free estimation of variance matrices. *Journal of Econometrics*, 181(2):165 – 180.

Bai, Z. and Silverstein, J. (2010). *Spectral Analysis of Large Dimensional Random Matrices*. Springer Series in Statistics, New York, 2 edition.

Bai, Z. D. and Silverstein, J. (1998). No eigenvalues outside the support of the limiting spectral distribution of large-dimensional sample covariance matrices. *The Annals of Probability*, 26(1):316–345.

Bai, Z. D. and Yin, Y. Q. (1993). Limit of the smallest eigenvalue of a large dimensional sample covariance matrix. *The Annals of Probability*, 21(3):1275–1294.

Bickel, P. J. and Levina, E. (2008a). Covariance regularization by thresholding. *The Annals of Statistics*, 36(6):2577–2604.

Bickel, P. J. and Levina, E. (2008b). Regularized estimation of large covariance matrices. *The Annals of Statistics*, 36(1):199–227.

Cai, T. T. and Zhou, H. H. (2012). Optimal rates of convergence for sparse covariance matrix estimation. *The Annals of Statistics*, 40(5):2389–2420.

Demiguel, V. and Nogales, F. J. (2009). A generalized approach to portfolio optimization: Improving performance by constraining portfolio norms. *Management Science*, 55(5):798–812.

Fan, J., Fan, Y., and Lv, J. (2008). High dimensional covariance matrix estimation using a factor model. *Journal of Econometrics*, 147(1):186–197.

Fan, J., Liao, Y., and Mincheva, M. (2011). High-dimensional covariance matrix estimation in approximate factor models. *The Annals of Statistics*, 39(6):3320–3356.

Fan, J., Liao, Y., and Mincheva, M. (2013). Large covariance estimation by thresholding principal orthogonal complements. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 75(4):603–680.

Friedman, J., Hastie, T., and Tibshirani, R. (2008). Sparse inverse covariance estimation with the graphical lasso. *Biostatistics*, 9(3):432–441.

James, W. and Stein, C. (1961). Estimation with quadratic loss. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*, pages 361–379, Berkeley, Calif. University of California Press.

Lam, C. and Fan, J. (2009). Sparsistency and rates of convergence in large covariance matrix estimation. *The Annals of Statistics*, 37(6B):4254–4278.

Lam, C. and Yao, Q. (2012). Factor modeling for high-dimensional time series: Inference for the number of factors. *The Annals of Statistics*, 40(2):694–726.

Lam, C., Yao, Q., and Bathia, N. (2011). Estimation of large latent factor models for time series data. *Biometrika*, 98(4):901–918.

Ledoit, O. and Péché, S. (2011). Eigenvectors of some large sample covariance matrix ensembles. *Probability Theory and Related Fields*, 151(1-2):233–264.

Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365 – 411.

Ledoit, O. and Wolf, M. (2012). Nonlinear shrinkage estimation of large-dimensional covariance matrices. *The Annals of Statistics*, 40(2):1024–1060.

Ledoit, O. and Wolf, M. (2013a). Optimal estimation of a large-dimensional covariance matrix under Stein's loss. ECON - Working Papers 122, Department of Economics - University of Zurich.

Ledoit, O. and Wolf, M. (2013b). Spectrum estimation: a unified framework for covariance matrix estimation and PCA in large dimensions. ECON - Working Papers 105, Department of Economics - University of Zurich.

Marčenko, V. and Pastur, L. (1967). Distribution of eigenvalues for some sets of random matrices. *Math. USSR-Sb*, 1:457–483.

Meinshausen, N. and Bühlmann, P. (2006). High-dimensional graphs and variable selection with the lasso. *The Annals of Statistics*, 34(3):1436–1462.

Pourahmadi, M. (2007). Cholesky decompositions and estimation of a covariance matrix: Orthogonality of variance-correlation parameters. *Biometrika*, 94(4):1006–1013.

Rothman, A. J., Levina, E., and Zhu, J. (2009). Generalized thresholding of large covariance matrices. *J. Amer. Statist. Assoc.*, 104(485):177–186.

Silverstein, J. and Bai, Z. (1995). On the empirical distribution of eigenvalues of a class of large dimensional random matrices. *Journal of Multivariate Analysis*, 54(2):175 – 192.

Silverstein, J. and Choi, S. (1995). Analysis of the limiting spectral distribution of large dimensional random matrices. *Journal of Multivariate Analysis*, 54(2):295 – 309.

Stein, C. (1975). Estimation of a covariance matrix. Rietz lecture, 39th Annual Meeting IMS. Atlanta, Georgia.

Stein, C. (1986). Lectures on the theory of estimation of many parameters. *Journal of Soviet Mathematics*, 34(1):1373–1403.

Stock, J. and Watson, M. (2005). Implications of dynamic factor models for var analysis. *NBER Working Papers*. No. 11467.

Won, J.-H., Lim, J., Kim, S.-J., and Rajaratnam, B. (2013). Condition-number-regularized covariance estimation. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 75(3):427–450.